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<u>Resumen/Summary</u> This technical report describes how to use the PANSA Computer Program, Version 5, and it includes examples to further clarify its use. The PANSA Program performs the stress and failure modes analysis of a rectangular flat sandwich panel with non corrugated core and equal or unequal thin faces, with different support conditions on the edges, subject to loads out of the plane and on its plane, both uniform or with a linear variation. The program checks if the EADS CASA failure criterion is met. The different failure modes analysed are: general buckling of the panel; <i>wrinkling</i>, <i>dimpling</i>, and <i>shear crimpling</i> on both faces, and stresses concentration on the upper face. This technical report substitutes the technical report NT-T-ADP-01039 User's Manual of the PANSA Program version 4.1 and corresponds to the version 5 of the program. The difference is the creation of a PC version of the program. To create this version, the input file has changed because now only one analysis can be done in each run. PROPIEDAD DE EADS CASA, Sociedad Unipersonal. Este documento no puede ser utilizado ni reproducido total o parcialmente sin la previa autorización escrita de la Dirección de Ingeniería y Tecnología de EADS-CASA. EADS CASA, Sociedad Unipersonal PROPERTY. This document shall neither be used nor completely or partially reproduced without previous written authorization by EADS-CASA Engineering and Technology Directorate.				
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1 Technical sheet of the program

Technical Note NT-T-ADP-08002

PROGRAM

PANSA

REFERENCEDATE

May 2000

OBJECTIVE

This program was designed for the analysis of strength and the different failure modes of a flat rectangular sandwich panel with non corrugated core and equal or unequal thin faces, with different support conditions on the edges, subject to out-of-plane loads and in-plane loads, both uniform or with a linear variation. The program checks if EADS CASA failure criterion is met.

The different failure modes analysed are: General Buckling, Wrinkling, Dimpling, Shear Crimping and Stress Concentration on the upper face.

LANGUAGE

Fortran 77

COMPUTER

SUN ENTERPRISE

NUMBER OF LINES (APROX.)

10.800

EXECUTION MODE

Interactive

AUTHOR

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VERSION

4.0

ADDITIONAL DOCUMENTS

Technical Note: NT-T-ADP-00005:

Manual Teórico del Programa PANSA, Versión 4.0

Technical Note: NT-T-ADP-00006:

Manual del Programador del Programa PANSA, Versión 4.0

2 Introduction

This program performs the strength and stability analysis of a flat rectangular sandwich panel with non-corrugated core and equal or unequal thin faces, with different support conditions on the edges and subject to normal loads and/or in-plane loads, both uniform or with a linear variation. The program checks if EADS CASA failure criterion is met.

The strength analysis checks the different failure modes that can appear on a sandwich panel:

- Wrinkling
- Dimpling
- Shear crimpling
- Stresses concentration on the outer face

This strength analysis is made following the procedure explained in the *Design Manual, Volume 11, Chapter 151*.

The way to perform the stress concentration analysis is explained in the section 2.1.

For the analysis of the sandwich panel stability, the program calculates the *critical general buckling load*. The critical buckling load of the panel is the loads combination, multiple or sub-multiple of the load applied, which creates the first instability of the panel. Therefore, the critical load is always homothetic with the load applied.

The program, based on the energetic method, has two application restrictions:

1. The problem to solve must be one with small strains, i.e., the maximum vertical displacement of the panel has to be lower than or equal to its total thickness.
2. The sandwich panel must have thin faces, so the distance between the midlines of the faces must be longer than six times the maximum panel thickness.

The method applied is based on the small strains theory for orthotropic sandwich panels developed by Libove and Batdorf, extending it to highly anisotropic panels.

The sandwich faces can be made of different materials (metal or composites) and can have different thickness. If they are made of composite materials, the program is valid for any type of laminate, symmetrically anisotropic or not. If they are made of metallic materials, these are assumed infinitely elastic.

The possible support conditions are: all edges simply supported, all edges clamped and edges with variable rotation stiffness. Whenever an edge is considered simply supported, it is possible to attribute a value to the rotation stiffness in order to consider a possible support mode that is intermediate between supported and clamped. When this value is equal to zero, the panel is simply supported. For isolated panels with stiffeners on their edges, this value can be taken as their torsional stiffness. (For more information, check the Theoretical Manual).

Finally, the results obtained by the program differ from the actual or exact buckling values in less than 1%. If the difference is higher, the program will warn about the maximum error produced.

2.1 Stress Concentration Analysis

The program performs the stress concentration analysis on the points selected by the user. The stress concentration is due to the action of the internal loads on that point, which are caused by the applied loads on the panel (including normal loads). The program analyses the stress distribution produced around a circular hole occurred on the selected point. In this analysis the following hypotheses are considered:

- Only *the upper face of the panel* is taken into account, because it is assumed that the hole is due to an external impact that could only occur on that face.
- The considered stresses concentration factors depend on the behaviour of the material of the sandwich face, which is assumed that it is *orthotropic*. To be coherent with this hypothesis, the face equivalent elastic modulus used is the one obtained only from the in-plane stiffness matrix, [A].
- The panel dimensions are assumed to be *very large* compared to the dimensions of the hole; so, the effects of the panel contour are negligible for the stresses distribution.

The following failure modes are considered under stresses concentration:

1. Fibre Failure

The load distribution along the hole contour is analysed considering only the points on one half of its perimeter (between 0 and 180°) every 5°. Only one half of the perimeter is considered because there are always loads symmetrical with regard to any diameter of the hole. The microstrains on the fibre directions on each point of the upper face are calculated*. These microstrains are compared with the allowable of the corresponding layer, and the critical value is that with the lowest reserve factor. This critical point is defined by its position on the hole contour (by the angle formed with the x-axis), by the orientation of the fibre where that minimum reserve factor is reached, and by the value of the microstrains and the components of the stresses vector on that point (on the x-y axes).

2. Matrix Failure

Now the load distribution is also analysed, on the hole contour, but taking the microstrains on the circumferential direction (the direction tangential to the contour). These microstrains are compared with the minimum value of the allowables of the layers (tension or compression, depending on the load sign), and finally the minimum value of the reserve factors obtained is given. The critical point is defined as in the previous case, but now the directions of the fibres are not considered and the microstrains are given for the circumferential direction.

3. Wrinkling Failure

The point on the semi perimeter of the hole where the maximum microstrains on the circumferential direction are located is considered. The components of the stresses vector are evaluated on this point. Using this vector, the failure due to the wrinkling is analysed. The point is defined as before, i.e., by its position on the contour (the angle regarding the x-axis), the microstrains on the circumferential direction and components of the stresses vector.

¹ If the properties of the faces are not defined from its stacking, but by its matrix [A] or by the equivalent moduli of the faces, -45°, 0°, 45° and 90° are supposed to be the fiber directions. On the other hand, if there is a fabric on a layer (this will be the assumption if the elastic moduli on directions 1 and 2 differ in less than 10%), the two corresponding normal directions are taken as the fiber directions.

2.2 Sign Criteria

The sign criteria are the usual: a positive flow means tension and a negative flow means compression, and the normal loads or pressures are positive when they are in the z -positive axis direction. Figure 1 shows a drawing of the applied loads and their signs. The values represented are positive.

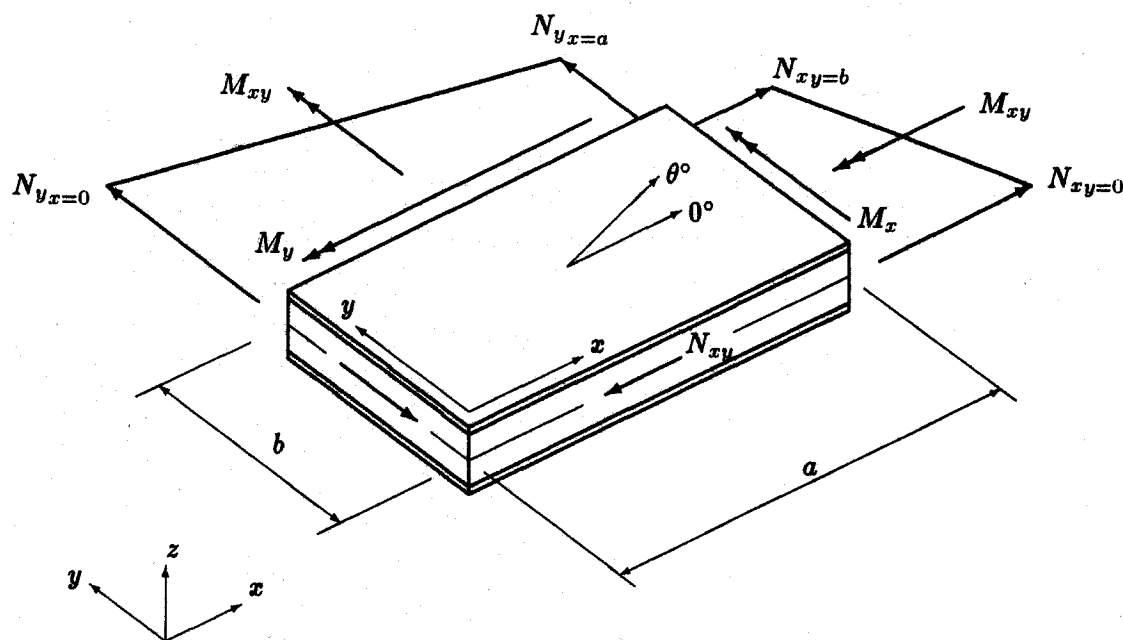


Figure 2.2.1: Applied flows and their signs.

In this version of the program, the x -axis of the panel should match the L (ribbon) direction, or the W direction of the core.

It is very important to pay attention to the shear sign, because its direction can notably influence on the results.

3 Program Execution

3.1 PC Server Version

The Visual Basic executable program (**pansa.exe**) is located in:

[\\gpr60005\InfoCalculo\Soft\Pansa](#)

In the folder [\\gpr60005\InfoCalculo\Soft\Pansa\Consola](#), there is another executable program, which is the Fortran program.

To execute the program, the user has to click twice on the Visual Basic executable program.

To ease this process, it is suggested creating a direct access to the program. To do so, drag with the mouse (do not copy) the **pansa.exe** program file to a directory (the Desktop, for example). In this case, to execute the program, the user just has to click twice on this shortcut.

3.2 Multivac Computer Version

To execute the PANSA program in the MULTIVAC Computer, a UNIX environment variable has been created with the name of the program. In order to execute this variable it is necessary that the user assigns the calculation symbols, which include the variables of all the available programs. The process is the following:

1. Enter the following lines in the **.profile** file:

```
ENV=$HOME/.kshrc
export ENV
BASH_ENV=$HOME/.bashrc
export BASH_ENV
```

2. If the user works with the **ksh** shell, enter the following line in the **.kshrc** file:

```
. asignaciones_calculo
```

3. If the user works with the **bash** shell, enter the following line in the **.bashrc** file:

```
. asignaciones_calculo
```

4. Type in the following command from the \$HOME directory:

```
./profile
```

(or exit and enter the SUN Computer again)

The executable is accessible from the SUN Computer *prompt* and from a command file. To execute the program, the '\$' symbol must be written, followed by the program name:

```
$pansa
```

As an input option, the input file name (complete) and the output file name (without extension) can be written following the program name:

```
$pansa input_file output_file
```

3.3 Execution Times

Basically, the execution time depends on: the support condition of the panel, the type of load and the dimensions of the panel.

The simulation of an intermediate support between simply supported and clamped also increases very significantly the execution time, which is longer when bigger is the stiffness considered, since it will require a higher number of terms for simulating by means of a development in sines, a type of strained geometry that looks more like a development in cosines because the value of the stiffness is very high.

With regard to the load type, shear loads need a much longer execution time than the compression loads.

On the other hand, the higher the rate of the sides, the higher the number of terms needed in the development of the strained geometry and, consequently, the higher will be the time of execution.

When different load cases should be analysed for the same panel, it is convenient to analyse them inside the same run, as a great deal of execution time can be saved because the calculation of the panel stiffness terms is not required to be repeated for each load case.

4 Data Input

4.1 PC Server Version

In the PC version, when the program starts, the following information window is displayed:

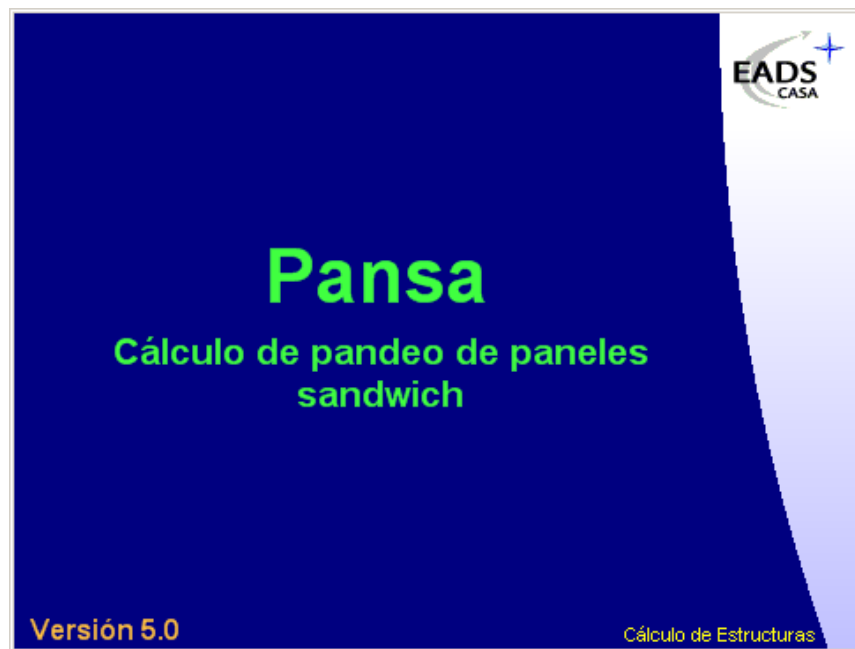


Figure 4.1. Program start

The input data can be entered via a file or using the graphical interface. If an input file is used, it is necessary to press the 'Open' button from the data form of the program and to choose the name of the file.

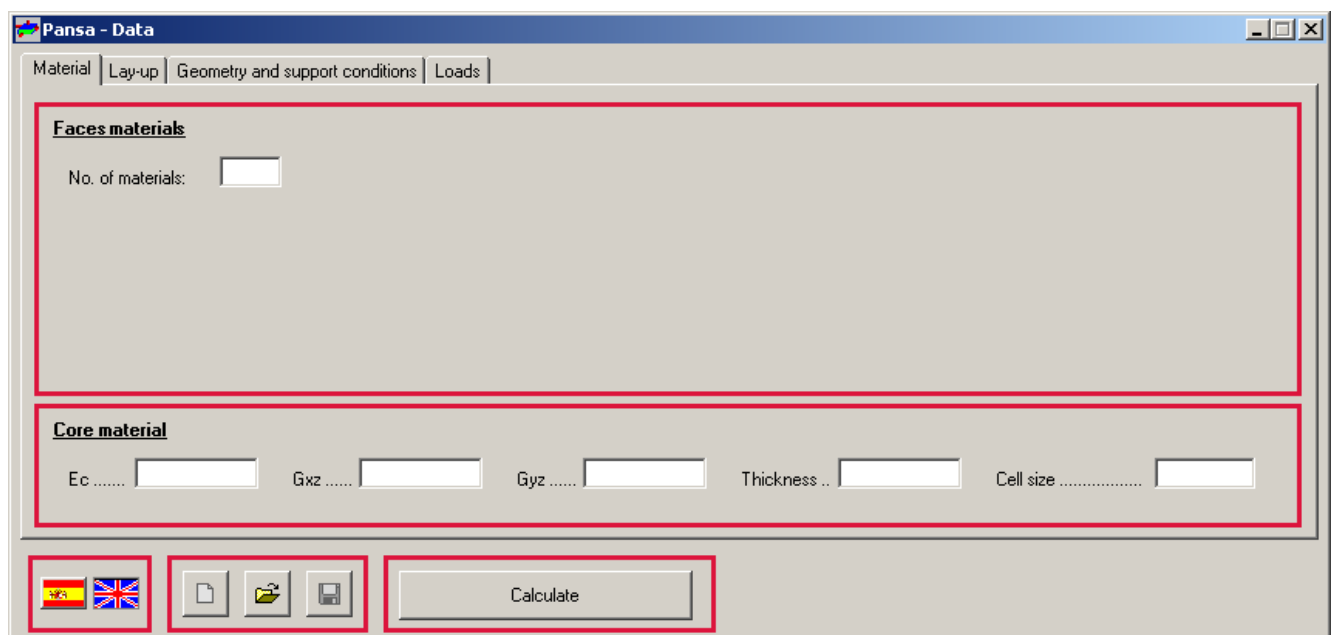


Figure 4.2. Input data

If the graphical interface is used, the user must fill in all the data required by pressing the different tabs. First, the materials number must be entered. Then a table appears and the boxes corresponding to each material property must be filled.

Pansa - Data

Material | Lay-up | Geometry and support conditions | Loads

Faces materials

No. of materials: 4

	E_1	E_2	G_{12}	μ	t	θ	$\mu \cdot 1t$	$\mu \cdot 1o$	E_{1B}	E_{2B}	G_{12B}
1	122000.0	10000.0	4700.0	0.30	0.125	0.0	12500	9500	109000.0	9000.0	4500.0
2	122000.0	10000.0	4700.0	0.30	0.125	45.0	12500	9500	109000.0	9000.0	4500.0
3	122000.0	10000.0	4700.0	0.30	0.125	-45.0	12500	9500	109000.0	9000.0	4500.0

Core material

E_c 193.0 G_{xz} 48.2 G_{yz} 24.1 Thickness .. 10.000 Cell size 4.8

Flags: [Spain] [UK] [File] [Folder] [Save] Calculate

Figure 4.3. Material data

When all data are entered, the user must press the 'Lay-up' tab and then choose the way to introduce the laminate data. If the 'using the stacking sequence' mode is chosen, the number of plies must be given for each face of the sandwich panel and another table appears where the user must choose the number of the material (using the order given before) of each ply. The thickness is calculated while the material of each ply is being selected.

Pansa - Data

Material | Lay-up | Geometry and support conditions | Loads

To define the laminate

☒ using the stacking sequence ☐ using stiffness matrix ☐ using elastic moduli ☐ Symmetrical

Upper face

No. of plies: 8 Thickness: 1.000

Ply	1	2	3	4	5	6	7
Material	2	3	1	2	3	4	3

Lower face

No. of plies: 6 Thickness: 0.750

Ply	1	2	3	4	5	6
Material	2	3	1	4	3	2

Flags: [Spain] [UK] [File] [Folder] [Save] Calculate

Figure 4.4. Laminate data using the stacking sequence

If the laminate is going to be defined by using the stiffness matrix, then the thickness, the allowables tension and compression strains in fibre direction and the mean values and allowable values of the A matrix must be entered. The stiffness matrices must be symmetrical.

Pansa - Data

Material | Lay-up | Geometry and support conditions | Loads

To define the laminate

☐ using the stacking sequence ☒ using stiffness matrix ☐ using elastic moduli

☒ Symmetrical

Upper face

Thickness: 0.750

μE_t 12500

μE_c 9500

Matrix A (mean values)

36350.8	15783.8	0.0
15783.8	36350.8	0.0
0.0	0.0	17042.1

Matrix A (allowables)

32650.9	13970.5	0.0
13970.5	32650.9	0.0
0.0	0.0	15305.4

Lower face

Thickness: 0.750

μE_t 12500

μE_c 9500

Matrix A (mean values)

36350.8	15783.8	0.0
15783.8	36350.8	0.0
0.0	0.0	17042.1

Matrix A (allowables)

32650.9	13970.5	0.0
13970.5	32650.9	0.0
0.0	0.0	15305.4

Calculate

Figure 4.5. Laminate data using the stiffness matrix

If the laminate is going to be defined by using the elastic modulus all the data required must be filled. In all cases, if 'Symmetrical' is chosen the data of the laminate of the lower face is copied from the data of the laminate of the upper face.

Pansa - Data

Material | Lay-up | Geometry and support conditions | Loads

To define the laminate

☐ using the stacking sequence ☐ using stiffness matrix ☒ using elastic moduli

☒ Symmetrical

Upper face

Thickness 0.750

E_1 (mean) 39329.9

E_2 (mean) 39329.9

G_{12} (mean) 22722.7

μ 0.430

E_1 (allw) 35564.3

E_2 (allw) 35564.3

G_{12} (allw) 20407.2

μE_t (allw) 12500

μE_c (allw) 9500

Lower face

Thickness 0.750

E_1 (mean) 39329.9

E_2 (mean) 39329.9

G_{12} (medio) 22722.7

μ 0.430

E_1 (allw) 35564.3

E_2 (allw) 35564.3

G_{12} (allw) 20407.2

μE_t (allw) 12500

μE_c (allw) 9500

Calculate

Figure 4.6. Laminate data using the elastic moduli

Third the 'Geometry and Support Conditions' tab must be clicked and the dimensions and the support conditions (simply supported or clamped) of the panel must be entered.

Pansa - Data

Material | Lay-up | **Geometry and support conditions** | Loads

Dimensions

Length 753.00 mm

Width 368.00 mm

Support conditions

☒ Simply supported

GJx 0.00 N/mm

GJy 0.00 N/mm

☐ Clamped

Spanish | English | [New] [Open] [Save] [Calculate]

Figure 4.7. Geometry and support conditions

Forth, the number of load cases is entered and then a different tab for each case appears and the user must fill in the in-plane loads. For the normal loads, pressure or concentrated loads can be chosen. The box on the right contains the information about the number and coordinates of the points where the analysis is required and if the maximum have to be calculated.

Pansa - Data

Material | Lay-up | Geometry and support conditions | **Loads**

No. of load cases: 2

Case 1 | Case 2

In-plane loads

Nx (y=0)	Nx (y=By)	Ny (x=0)	Ny (x=Ax)	Nxy
-1.0	-1.0	0.0	0.0	0.0

Normal to the panel plane loads

☒ Pressure loads

Pressure at x=0 y=0	-0.00688	Pressure at x=Ax y=By ...	-0.00344
Pressure at x=Ax y=0	-0.00344	Pressure at x=0 y=By	-0.00688

☐ Concentrated loads

Calculation of maximums?

☐ Yes ☒ No

No. of points: 1

	x	y
1	376.500	184.000

Spanish | English | [New] [Open] [Save] [Calculate]

Figure 4.8. Loads data

Then the user can save the data pressing 'Save' button or press the 'Calculation' button and wait for the results. If the 'Save' button is pressed, the program creates a data file with a '. DAT' extension that contains the data used in the analysis, properly sorted. This file can be used as the data source to repeat the same analysis, or may be used as a basis for a new data file to perform a new analysis.

4.2 Multivac Computer Version

Data input is performed by means of a data file. The dialog with the program is the following:

<u>Program PANPA</u>	
Calculation of the allowables of linear buckling (small strains) of a panel with the following characteristics:	
- Material : Metallic or anisotropic laminate - Geometry : Plane, rectangular, constant thickness, stiffened or not in two directions - Supports : 4 sides simply supported Sides // X clamped, sides // Y supported Sides // X supported, sides // Y clamped 4 sides clamped - Loads : Any combination of uniform loads with linear variation in its plane	
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To continue, press <RETURN>

Notes

- In case of a metallic bending panel, the procedure used in this program to study plastic reduction of buckling allowables is only approximated, so errors can be obtained, the greater the reduction, the greater the errors.
- This software and the data that are produced by it are EADS-CASA property.
The total or partial reproduction of this software is forbidden, as well as its use out of the EADS-CASA environment.

To continue, press <RETURN>

The input data file must be introduced interactively via the keyboard:

Name of the input data file →
--

The full name of the file containing the output data must be given.

Name of the output file without extension [PANSA] →

4.3 Input Data File

The composition of the input data file is the following:

```

N_m
1, E1(1), E2(1), G(1), μ(1), t(1), θ(1), εt(1), εc(1), EB1(1), EB2(1), GB(1)
2, E1(2), E2(2), G(2), μ(2), t(2), θ(2), εt(2), εc(2), EB1(2), EB2(2), GB(2)
- ----
i, E1(i), E2(i), G(i), μ(i), t(i), θ(i), εt(i), εc(i), EB1(i), EB2(i), GB(i)
- ----
- ----
N_m, E1(N_m), E2(N_m), G(N_m), μ(N_m), t(N_m), θ(N_m), εt(N_m), εc(N_m), EB1(N_m), EB2(N_m), GB(N_m)
Stacking_seq
Num_plies_upper face or Thickness upper face
Stacking code or Terms of A matrix or Elastic moduli of the upper face (mean values)
(1) Terms of A matrix or Elastic moduli, Et, Ec of the upper face (allowable values)
S or D
(2) Num_plies_lower face or Thickness lower face
(2) Stacking code or Terms of A matrix or Elastic moduli of the lower face (mean values)
(2) Terms of A matrix or Elastic moduli, Et, Ec of the lower face (allowable values)
Ec, Gxz, Gyz, core_thickness, cell_size
Ax, By
Support_cond, GJx, GJy
Iden_panel
Num_load_cases
Nx_y=0, Nx_y=b, Ny_x=0, Ny_x=a, Nxy
Code, P1, P2, P3, P4
or: Code,nloads,f1,xf1yf1,...,fi,xfi,yfi
or: Code
} Num_load_cases
times
N_points [,x_p_1, y_p_1, ..., x_p_i, y_p_i]
Maximums

```

(1) This line doesn't exist if the stacking sequence type chosen is 0.

Data corresponding to the sandwich lower face must be only introduced if this face is different from the upper face. The two faces are equal if the sandwich is symmetrical with regards to its midline. The upper face is the one situated in the z-positive axis (or the face with the higher Z in any cases).

The symbols used are:

N_m

Number of materials whose properties will be given next. A different material is a material that presents any data different from the previous one, including the orientation of the fibre (θ). It is an integer value.

Next, the properties of each of the N_m materials are given. One line per material is written with the following data:

- i** Identification number of the material.
- E₁(i)** Longitudinal elastic modulus of the 'i' material (mean value).
- E₂(i)** Transverse elastic modulus of the 'i' material (mean value).
- G(i)** Shear modulus of the 'i' material (mean value).
- μ(i)** Poisson's ratio of the 'i' material (mean value).
- t(i)** Ply thickness of the 'i' material.
- θ(i)** Ply orientation angle of the 'i' material regarding to the panel x-axis.
- ε_t(i)** Allowable tension strain of the 'i' material in the fibre direction.
- ε_c(i)** Allowable compression strain of the 'i' material in the fibre direction.
- E_{B1}(i)** Longitudinal elastic modulus of the 'i' material (B value).
- E_{B2}(i)** Transverse elastic modulus of the 'i' material (B value).
- G_B(i)** Shear modulus of the 'i' material (B value).

The last three are only used for the calculation of the panel stability.

Stacking_seq

Type of stacking sequence of the faces. It can have the following values:

- 0** Indicates that the real sequence will be given in the stacking code, i.e., a sequence of integer numbers that indicate the material of each layer of the corresponding face.
- 1** Indicates that the in-plane stiffness matrix (A Matrix) of the corresponding face will be given in the stacking code. It is assumed that the face is symmetrical.
- 2** Indicates that the elastic modulus on the material plane of the corresponding face will be given in the stacking code. In this case, it is assumed that the material is orthotropic.

The following values depend on the type of stacking sequence chosen:

- If **Stacking_seq** = 0, then two lines are entered:

Num_plyes_upper_face

Number of layers of the laminate of the upper face (the face with maximum Z). Integer value.

Stacking code of the upper face

A sequence of integer numbers that indicate the type of material of each layer. The order to enter the laminate layers is always from the outside to the inside of the sandwich, that is, from higher to lower Z.

- If **Stacking_seq** = 1, then three lines are entered:

Thickness_upper_face

Thickness of the laminate that forms the upper face. It is a real value.

Terms of the stiffness matrix (mean values)

The terms of the in-plane stiffness matrix, A Matrix, of the laminate that forms the upper face calculated from the average properties of the materials, in the following order:

$A(1,1), A(1,2), A(1,3), A(2,2), A(2,3), A(3,3)$

Terms of the stiffness matrix, ϵ_t y ϵ_c (allowable values)

The terms of the in-plane stiffness matrix (A Matrix) calculated from the B values (allowable values) of the laminate material that forms the upper face and the allowable tension and compression strains in this order:

$A(1,1), A(1,2), A(1,3), A(2,2), A(2,3), A(3,3), \epsilon_t, \epsilon_c$

In this case, to calculate the rest of the mechanical properties of this face, it is assumed that the matrix is symmetrical.

- If **Stacking_seq** = 2, three lines are entered:

Thickness_upper_face

Thickness of the laminate that forms the upper face. It is a real value.

Elastic moduli of the upper face (mean values)

The mean values of the elastic moduli on the x and y directions, the shear modulus and the Poisson modulus of the upper face material are given in the following order:

$E_{xmean}, E_{ymean}, G_{xy}, \mu_{xy}$

Elastic moduli of the upper face, ϵ_t y ϵ_c (allowable values)

The B values (allowables) of the elastic moduli on the x and y directions, the shear modulus and the allowable tension and compression strains of the material of the upper face are given in the following order:

E_{xallow} , E_{yallow} , $G_{xyallow}$, ϵ_c , ϵ_t

In this case, to calculate the rest of the mechanical properties of this face, it is assumed that the face is orthotropic.

S or D

It can have two values: s (symmetrical) or d (different). These letters can be upper or lower case.

The s or S value indicates that both faces are equal and therefore, the sandwich is symmetrical with regard to its midline. Lines indicate with (2) in the previous text box are not required.

The d or D value indicates that both faces are not equal and the definition of the lower face must be entered, in the same way as for the upper face. In this case, the order used to enter the laminate layers is also outside to the inside of the sandwich, but here it is from the lower to the higher Z .

Ec, Gxz, Gyz, core_thickness, cell_size

They are the properties of the core: the compression elastic modulus (z direction), the transverse shear moduli on the xz plane and the yz plane, the thickness and the cell size, respectively.

Ax, By

They are the dimensions of the sandwich analysed, according to the x and y directions, respectively.

Support, GJx, GJy

They are three numbers, the first is an integer and the other two are real, that indicate the support condition of the sandwich panel. **Support** can only have the following two values:

- 1 Simply supported edges.
- 4 Clamped edges.

GJx, **GJy** are the rotation stiffnesses (GJ) of the sides parallel to x and y , respectively. It allows analysing intermediate edge conditions between simply supported and clamped edges. So, if both values are equal to 0, it is a simply supported panel, while for very big values, the support condition is like a clamping. For isolated panels with stiffeners on their edges, these values can be taken as their torsional stiffness.

Iden_panel

Alphanumeric variable (132 characters maximum) used for panel identification.

Num_load_cases

It is an integer value that indicates the number of load cases to be analysed for the current configuration. A maximum of 20 cases can be analysed. Two lines must be written for each load case: one with the in-plane loads and one with the normal loads.

$Nx_y=0$, $Nx_y=b$, $Ny_x=0$, $Ny_x=a$, Nxy

They are the applied flows: on the x direction (on points $y = 0$ and $y = B_y$, respectively), on the y direction (on points $x = 0$ and $x = A_x$, respectively) and the shear flow.

In the case of a linear variation of the applied load with a value equal to zero, on one of the ends, that end should not be located on the $x = 0$ axis or $y = 0$ axis. If this occurred, the program changes automatically the position of the origin; so the value of the stress applied on the axis is not null, which does not change the nature of the problem, which is exactly the same. The reason for this is that the value of the stress applied on the axis is taken as a reference to define each variation along that edge, so it must not be zero.

Code

Integer number that indicates the normal load or pressure applied to the panel in combination with the in-plane loads indicated in the previous line in the corresponding load case. The following values can be taken:

- 0** Indicates that this load case has no out-of-plane loads. No other data will be entered on that line.
- 1** Indicates that the panel is subjected to a pressure loads whose values on the four corners of the panel will be entered next: **P_i** ($i = 1$ to 4), beginning with the corner corresponding to the coordinates origin (0,0), and following by the coordinates corner (A_x , 0), (A_x , B_y) and finally (0, B_y). It is assumed that the pressure variation is linear along the axis between those points.
- 2** Indicates that next, a series of normal concentrated loads on the panel and their application points will be given. **$Nloads$** is the number of loads to be entered, **f_i** is the value of 'i' load and **xf_i** and **yf_i** are the coordinates of the application point of the 'i' load ($i = 1$ to **$Nloads$**). **$Nloads$** can have a maximum value of 20. If all the previous data do not fit on one line, it is possible to continue on the following line, by ending the previous line with a comma and a '+' sign.

N_points

It is the number of points on the panel where the results must be obtained. **x_p_i** and **y_p_i** are the coordinates of the points to analyse (1 to **N_points**) (20 points maximum). If a '0' is entered here the program only makes the buckling analysis.

Maximums

It can have two values: **si (yes)** or **no** . If the user chooses **si** , the program will calculate the points where are located the maximum values of the variables vertical displacement w , bending moments M_x and M_y , torsional moment M_{xy} and shear loads Q_x and Q_y . If the user chooses **no** , the program does not calculate these points. Only the first letter of each value is required, it can be in upper or lower case (**s** or **n** or **S** or **N**).

5 Results and Output files

The results output is presented in two files:

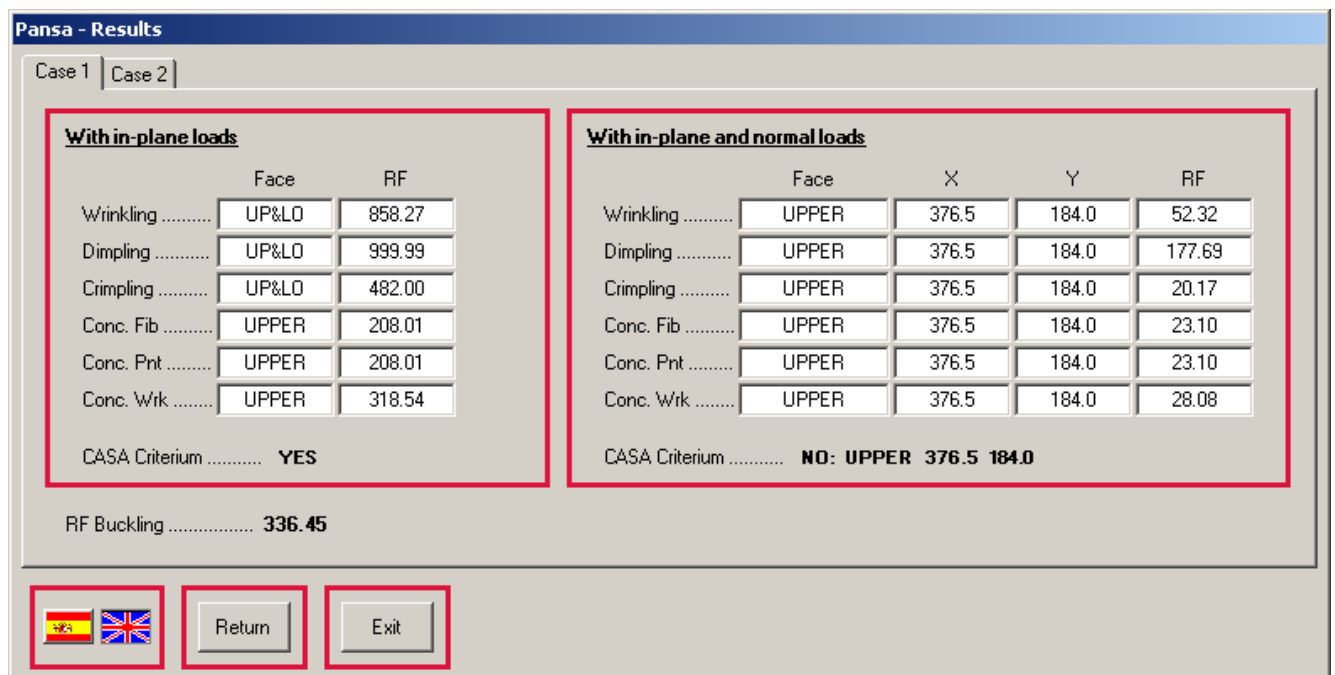
<name>.RES and **<name>.AUX**

where ‘<name>’ is the name of the output file given to the program.

The analysis results are in the ‘.RES’ file, whereas the ‘.AUX’ file contains additional or auxiliary information.

5.1 PC Server Version

In the PC version, when all data is entered and the button ‘Calculate’ is pressed, the program runs and two results files are created. Then a form with the results calculated by the program is shown:



Pansa - Results

Case 1 | Case 2

With in-plane loads		
	Face	RF
Wrinkling	UP&LO	858.27
Dimpling	UP&LO	999.99
Crimpling	UP&LO	482.00
Conc. Fib	UPPER	208.01
Conc. Pnt	UPPER	208.01
Conc. Wtk	UPPER	318.54
CASA Criterium YES		

With in-plane and normal loads				
	Face	X	Y	RF
Wrinkling	UPPER	376.5	184.0	52.32
Dimpling	UPPER	376.5	184.0	177.69
Crimpling	UPPER	376.5	184.0	20.17
Conc. Fib	UPPER	376.5	184.0	23.10
Conc. Pnt	UPPER	376.5	184.0	23.10
Conc. Wtk	UPPER	376.5	184.0	28.08
CASA Criterium NO: UPPER 376.5 184.0				

RF Buckling **336.45**






Figure 5.1. Results

For each load case a summary of the results that the files created contain are shown. At the bottom the minimum reserve factor and the load case that produces it are written.

Here the user can return to the data or finish the program.

5.2 Multivac Computer Version

Every time the program is run, it generates the two files mentioned before:

<Name>.RES and <Name>.AUX

where <Name> is the filename that was entered when the program requested a name for the output files.

5.3 Output Files

The output '.RES' file is structured in several groups:

1. *Sandwich properties*

- Stacking sequence of the sandwich faces. For each layer are given:
 - Layer number.
 - Distance from the layer midline to the sandwich midline.
 - Mean and allowable values of the elastic moduli (directions 1 and 2) and the shear modulus.
 - Thickness and orientation.
 - Allowable tension and compression microstrains.
- Core properties: thickness, distance to the midline and compression and shear moduli.
- Faces properties (mean values). For each face are given:
 - In-plane stiffness matrix [A].
 - Equivalent elastic on x direction and on y direction moduli and shear modulus.
 - Poisson's ratio.
- Sandwich properties (mean values):
 - Stiffness matrices: $[A]_{\text{mean}}$, $[B]_{\text{mean}}$, $[D]_{\text{mean}}$.
 - Equivalent elastic x direction and y direction moduli ($E_{x\text{mean}}$, $E_{y\text{mean}}$) and shear modulus ($E_{xy\text{mean}}$) calculated from the average properties.
 - Equivalent Poisson moduli on the x direction (μ_x) and y direction (μ_y), obtained from the average properties.
 - Bending stiffnesses of the panel per width unit $D_{x\text{mean}}$, $D_{y\text{mean}}$ and $D_{xy\text{mean}}$ (also called stiffnesses in sandwich notation or stiffnesses in especially orthotropic plate terms), calculated from the average properties:

$$D_x = D_{11} * (1 - \beta_x \beta_y) \quad D_y = D_{22} * (1 - \beta_x \beta_y) \quad D_{xy} = D_{66}$$

with $\beta_x = D_{12} / D_{22}$ and $\beta_y = D_{12} / D_{11}$ being the apparent Poisson moduli defined as a function of the curvatures. Remember that these stiffnesses link, in orthotropic materials, the curvatures with the internal moments.

- Sandwich properties (B values):
 - Stiffness matrices: $[A]_{allow}$, $[B]_{allow}$, $[D]_{allow}$.
 - Equivalent elastic x direction and y direction moduli (E_{xallow} , E_{yallow}) and shear modulus ($E_{xyallow}$) calculated from the allowable properties.
 - Bending stiffnesses of the panel per width unit D_{xallow} , D_{yallow} and $D_{xyallow}$, calculated from the allowable properties.

2. Geometric data of the panel

- Dimensions of the panel on x and y directions.
- Thickness: faces, core and total thicknesses.
- Cell size of the core.
- Support conditions.
- Panel especially orthotropic or not. A panel is especially orthotropic if the following equations are true:

$$A_{13} = A_{23} = A_{31} = A_{32} = 0$$

$$B_{13} = B_{23} = B_{31} = B_{32} = 0$$

$$D_{13} = D_{23} = D_{31} = D_{32} = 0$$

3. Local analysis for each load case

- Load case number.
- In-plane applied flows and applied pressures for that load case. If the behaviour of the panel under this load case is not of small displacements, a message indicating so is displayed clearly. In this case, the membrane effects are not negligible with regard to the bending effects, and the results obtained for this load case are not valid, although they are given on the following pages.
- Results on the analysed points. For each point where the analysis was requested, and for each point where the maximum of w , M_x , M_y , M_{xy} , Q_x and Q_y are located (only if the calculation of maximum values was requested in the data input file), the following is displayed:
 - Coordinates of the point.
 - Internal loads at that point: the three flows (N_x , N_y and N_{xy}), the three moments per length unit (M_x , M_y and M_{xy}) and the components of transverse shear flow per length unit (Q_x and Q_y).
 - The vertical displacement on the analysed point (w).
 - The midplane strains on the point.
 - The strains on the layer axis (fibre 1 and matrix 2) of each layer.

- The strains on the panel axis (x and y) of each layer.
- The stresses on the layer axis (fibre 1 and matrix 2) of each layer.
- The stresses on the panel axis (x and y) of each layer.
- The stresses on the main axis of each layer.

The stresses and strains output of each layer is only given when the configuration of the faces has been performed by means of the stacking code and not by means of the in-plane stiffness matrix or equivalent elastic moduli. In this last case, the mean stresses and strains on the core and on both faces are given.

- Maximum stresses and strains due to concentration: the maximum stress vector, the maximum strain and the angle where this maximum occurred are given for the three failure modes considered in the stresses concentration analysis.
- Summary of local results: reserve factors and allowables are given on the point that is being analysed for the current load case (including the normal loads). The two panel faces are evaluated, except for the stresses concentration, which is considered only on the upper face. The results given depends on the failure mode analysed, that is:

1. *Wrinkling*

F_{cw} : Wrinkling compression allowable stress of the panel.

F_{sw} : Wrinkling allowable stress of the panel subject to an in-plane shear load.

RF: Minimum reserve factor of the interaction of the applied loads.

2. *Dimpling*

F_{cd} : Dimpling compression allowable stress of the panel.

F_{sd} : Dimpling allowable stress of the panel subject to an in-plane shear load.

RF: Minimum reserve factor of the interaction of the applied loads.

3. *Shear Crimpling*

F_{sc} : Shear crimpling allowable stress of the panel.

RF: Reserve factor.

4. *Stresses concentration**

RF (*fibre*): Minimum reserve factor on the fibre directions.

RF (*point*): Minimum reserve factor on any direction, whether it matches or not the fibre direction.

RF (*Wrin*): Wrinkling reserve factor corresponding to the stresses for maximum circumferential strains.

* The stresses concentration factors, and the reserve factors obtained for concentration can be non-conservative if the sandwich faces are not orthotropic.

4. Panel stability

- Buckling results for each load case:
 - In-plane applied flows and critical buckling flow. If the buckling eigenvalues could not be calculated, 0.0 is written here and a warning message is given in the .AUX file.
 - Number of semi waves on each direction (N_{ox} , N_{oy}).
 - Buckling reserve factor.
- Minimum reserve factor among all the load cases analysed where the eigenvalues calculation converged.

5. Summary of allowables and reserve factors (only under in-plane loads) for each load case calculated on the centre of the panel

- Maximum stresses and strain due to concentration on the centre of the panel: the maximum stress vector, the maximum strain and the angle where this maximum occurred are given for the three failure modes considered in the stresses concentration analysis, considering only in-plane loads.
- Reserve and allowable factors considering only in-plane loads on the centre of the panel: the load is distributed between the panel faces, according to its relative stiffness, and the results are evaluated on both faces, except for the stresses concentration that is only considered on the upper face. The same failure modes as in group data 3 are considered, adding the buckling reserve factor of group data 4.

6. Summary of reserve factors for each load case and for each panel

- Reserve factors due to buckling and to the different failure modes, considering only in-plane loads and calculated on the centre of the panel.
- Minimum reserve factors with normal loads for the failure modes. The minimum RF from those calculated on all the points where the analysis has been done, the point coordinates of the point where this minimum occurs and the (upper or lower) panel face where the point is located are indicated.
- CASA EADS failure criterion evaluated with and without normal loads applied. CASA EADS failure criterion is fulfilled when the reserve factors due to buckling, wrinkling, dimpling and shear crimping are, at least, 1.5 times higher than the reserve factor due to concentration on the fibre direction. When the normal loads are considered, CASA EADS criterion is evaluated on each point of the analysed panels, and it is fulfilled when it is met for all of them. In case it is not fulfilled, the points that originate it are indicated.

Data groups 3, 5 and 6 are included only if any point of calculation was given.

5.4 Auxiliary Output File

The .AUX output file indicates the additional information about calculations performed, with some interesting data.

For each load case analysed the following information is given:

- In the calculation of the strained geometry under normal loads and for each iteration performed until the convergence is reached, are given:
 - The number of terms of the development.
 - The vertical displacement on the centre of the panel.
 - The midplane curvatures on the centre of the panel.
 - The midplane bending curvatures on the centre of the panel.
 - The transverse shear flows per length unit.

That information only appears if any point of calculation was given in the input file.

- In the calculation of the buckling eigenvalues, the following information is given:
 - The number of iterations until the convergence is met.
 - The relative difference or maximum error of the reserve factor obtained.
 - The number of terms of the development in each iteration.
 - The buckling reserve factor calculated in each iteration.
 - The reserve factors calculated after the iterations for the first five buckling modes.
 - The semi waves on the buckling strained for the first five buckling modes.

5.5 Warning Messages

The program warnings are the following:

1. During the opening of the input file:

* FILE NOT FOUND OR WRITE-PROTECTED

The program requests again the name of the data file.

2. During the reading of the input data:

During the data reading, the program checks the value of the most important data. The possible warnings are:

***** ERROR IN THE DATA ENTRY *****

***** MAXIMUM MATERIALS NUMBER EXCEDEED ***

***** ERROR ENTERING THE MATERIALS PROPERTIES ***

***** ERROR IN THE TYPE OF STACKING SEQUENCE ***

***** ERROR IN THE DEFINITION OF THE UPPER FACE LAY-UP ***

***** ERROR IN THE DEFINITION OF THE LOWER FACE LAY-UP *****

***** ERROR ENTERING THE CORE PROPERTIES *****

***** ERROR ENTERING THE PANEL DIMENSIONS *****

INFINITE LONG PANEL: DECREASE THE SIDES RATIO TO SAVE EXECUTION TIME

***** ERROR ENTERING THE SUPPORT CONDITIONS *****

***** INCORRECT PANEL IDENTIFICATION NAME *****

***** ERROR ENTERING THE LOAD CASES NUMBER *****

***** THE MAXIMUM NUMBER OF LOAD CASES IS 20 *****

***** ERROR ENTERING THE IN PLANE-LOADS *****

***** ERROR ENTERING THE TYPE OF NORMAL LOADS *****

***** ERROR ENTERING THE PRESSURE LOADS *****

***** ERROR ENTERING THE CONCENTRATED LOADS *****

***** ERROR ENTERING THE POINTS TO ANALYZE *****

***** THE MAXIMUM NUMBER OF POINTS TO ANALYZE IS 20 *****

If the program detects one of these errors during the data input, it shows the corresponding message on screen and stops the program execution.

3. Program applicability:

When, after reading the input data, the program checks their applicability the following messages might be shown:

* THIS SANDWICH PANEL DOES NOT HAVE A THIN WALL !

The theory behind the program limits its applicability to a sandwich made of thin faces, i.e., being the distance between the midlines of the upper faces six times the thickest one. Therefore, if this warning appears, the panel given data cannot be analysed by this program.

4. During the execution of the calculations:

The following warnings indicate error messages found by subroutines generally used for matrix calculations. They are considered on these subroutines and they are not displayed during the normal execution of the program:

**** Error en la subrutina SOLAIJ:
La matriz para el sistema de ecuaciones está mal
condicionada.

**** Error en la subrutina SOLAIJ:
La matriz para el sistema de ecuaciones no es simétrica.

**** Error en la subrutina SOLECU:
La matriz para el sistema de ecuaciones está mal
condicionada.

**** Error en la subrutina SOLECU:
La matriz para el sistema de ecuaciones no es simétrica.

If any of these warnings is displayed the execution is interrupted, and you should call the Methods Department.

5. In the .AUX file:

The program shows a warning if the maximum number of development terms allowed is reached. The given result corresponds to the maximum number of development terms allowed but the convergence has not been necessarily reached.

* THE MAXIMUM NUMBER OF DEVELOPMENT TERMS WAS REACHED

The program also shows a warning in the .AUX output file if the last iteration to reach the convergence failed. The given results correspond to the previous iteration, but the convergence has not been necessarily reached.

* THE LAST ITERATION FAILED

Warnings related to the *buckling eigenvalues calculation* process can also be displayed. For the cases where the process did not converge or the result is not precise enough, the following warning is displayed on screen, followed by a remark with additional information about the error causes:

*** ERROR IN THE BUCKLING EIGENVALUES CALCULATION**

This warning is also included in the .AUX output file, in the corresponding section. The .RES file does not identify the error and writes a null value for the buckling reserve factor.

6. In the .RES file:

These warnings are written in the .RES output file and they are related to the applicability of the concentration failure criterion and to the program hypothesis of only considering small displacements.

* STRESS CONCENTRATION FAILURE CRITERIUM COULD BE NO
CONSERVATIVE DUE TO THE SANDWICH FACES ARE NO ORTOTROPIC

* !!! PANEL BEHAVIOUR FOR CURRENT LOAD CASE DOES NOT MATCH THE
SMALL DISPLACEMENTS HYPOTHESIS ($w > t$).
RESULTS FOR THIS LOAD CASE ARE NOT VALID !!!

6 Differences with the Previous Versions

- **Version 2.0 → Version 3.0:**

- (1) Adaptation of the original program compiled with the VAX/VMS computer to allow its the compilation under the SUN UNIX operating system.
- (2) Change from the old IMSL subroutines to the latest available version.
- (3) An evaluation of an index for the eigenvalues calculation (*performance index*) is included, which gives an idea of the precision obtained during the calculation. If this index shows a low precision, the calculation is considered incorrect and an error is displayed during the output.

- **Version 3.0 → Version 4.0:**

- *Differences in data input*

- (1) The application of normal loads is allowed; this data should be given in a series of points on the panel surface.
- (2) The stress analysis on the panel under normal and in-plane loads that was performed by the ARSA program is integrated, so the user must indicate the points to perform this analysis.

- *Differences in the results output*

- (3) The calculation of reserve factors due to the sandwich failure modes (wrinkling, crimping, dimpling and stresses concentration) is added, considering normal loads on the points defined by the user.
- (4) The application of the stress concentration failure criterion is limited to the upper (external) face of the panel, because it is assumed that the impacts can only occur on that face.
- (5) In-plane applied loads are distributed between the faces proportionally to its stiffness.
- (6) When the elastic modulus of a layer on the fibre direction is different from the one on its normal direction in less than 10%, the material of this layer is considered a fabric in the stresses concentration calculation.
- (7) The parallel axes theorem formulation, for the stiffness matrices calculation from the average allowable properties of the faces, has been corrected.
- (8) The calculation of the equivalent elastic modulus of the faces has been corrected. Now it is obtained from the inverse of the complete sandwich stiffness matrix ([A], [B], [D]). The orthotropic behaviour hypothesis is maintained in the calculation of the stresses concentration factor.
- (9) CASA EADS failure criterion has been modified for sandwich panels. In the new criterion formulation the reserve factors due to failure modes need to be at least 1.5 times higher than the reserve factor due to stress concentration. The criterion is evaluated on each of the points analysed under normal loads (it must be met for all of them) and on the global panel considering only the in-plane loads.
- (10) A summary, where the minimum reserve factors of all the points analysed with normal loads (for the failure modes: *wrinkling*, *dimpling*, *crimping* and concentration) are displayed, is included.

- (11) The results of allowables and reserve factors for each failure mode and for both faces of the panel are presented separately.
- (12) In the data group of the output file, along with the panel properties, the stiffness matrices of each of the faces and their elastic moduli are included.
- (13) In the data group of the output file, along with the failure allowables due to concentration, stresses, strains and the corresponding fibre direction are included.

- **Version 4.0 → Version 4.1:**

- *Differences in data input*

- (1) The support conditions of the sandwich can be simply supported or clamped or a situation between the two previous situations. In the version 4.0, if the load case included normal loads the panel couldn't be clamped, although the program calculated on the right way the buckling of a clamped panel.

- **Version 4.1 → Version 5.0:**

- (1) New PC version.
 - (2) Differences in data input: only one analysis is done for each input file, so the number of analysis is removed from the data written in the input file.

7 Applicability and Program Limitations

The most important program limitations were described in the purpose of this program:

- It is a linear problem of small strains, so the maximum vertical displacement should be lower than or equal to the thickness of the panel to be analysed.
- The sandwich panels should have thin faces, which means that the distance between the midlines of the faces should be more than six times the thickest one.
- The stresses concentration factors, and therefore their corresponding reserve factors due to concentration failure can be non-conservative if the sandwich faces are not orthotropic. CASA EADS criterion also depends on these values and might be non-conservative.

An important condition in this version is that the direction of the x -axis of the panel should match either the L direction (ribbon) or the W direction of the core, which means that the panel edges must be parallel to these two core directions.

On the other hand, there are several limitations regarding to the parameters dimensions, so they can be easily avoided. They are the following:

- The maximum number of possible different materials is 50.
- The maximum number of layers of a face is 100.
- The maximum number of load cases of each analysis is 20.
- The maximum number of points to analyse is 20.

The maximum number of terms to be taken in the development of the strained geometry in x direction and in y direction is such that their product is 225, each being proportional to the length of its side and being always lower than 20. This number is high enough, as to guarantee in almost every case an error below 1% with regard to the exact value. When this limitation is exceeded, the program warns about the maximum error incurred.

8 Examples

Next, six examples are presented to clarify the different possibilities of the program.

8.1 Example 1

The first panel is rectangular, 753 mm length, 368 mm width, symmetrical. Its faces are made of a composite material laminate of 6 layers, from the outside to the inside (45/-45/0/90/-45/45). It is simply supported and subject to two load cases:

1. A uniform compression flow on the x direction of 1 N/mm, and a uniform pressure of 6881 N/m² oriented on the z -negative direction.
2. A positive shear flow of 1 N/mm without out-of-plane load.

The analysis is wanted on the centre of the panel.

The **core properties** are:

$$\begin{aligned} E_c &= 193 \text{ N/mm}^2 & G_{xz} &= 48.2 \text{ N/mm}^2 & G_{yz} &= 24.1 \text{ N/mm}^2 \\ t &= 10 \text{ mm} & s &= 4.8 \text{ mm} \end{aligned}$$

The **mechanical properties** of the material of the faces are:

Average properties:

$$\begin{aligned} E_x &= 122000 \text{ N/mm}^2 & E_y &= 10000 \text{ N/mm}^2 & G &= 4700 \text{ N/mm}^2 \\ \mu &= 0.3 & t_n &= 0.125 \text{ mm} \end{aligned}$$

Allowable properties, (B values):

$$\begin{aligned} E_x &= 109000 \text{ N/mm}^2 & E_y &= 9000 \text{ N/mm}^2 & G &= 4500 \text{ N/mm}^2 \\ \epsilon_t &= 12500\text{E-}06 \text{ mm/mm} & \epsilon_c &= 9500\text{E-}06 \text{ mm/mm} \end{aligned}$$

The data input file is on the following page.

ej_pansa_1.dat

```
! Materials
4
1,122000,10000,4700,.3,.125,0,12500E-6,9500E-6,109000,9000,4500
2,122000,10000,4700,.3,.125,45,12500E-6,9500E-6,109000,9000,4500
3,122000,10000,4700,.3,.125,-45,12500E-6,9500E-6,109000,9000,4500
4,122000,10000,4700,.3,.125,90,12500E-6,9500E-6,109000,9000,4500
! Laminate
0
6
2,3,1,4,3,2
S
! Core properties
193.0,48.2,24.1,10.0,4.8
! Panel dimensions
753,368
! Support conditions
1,0.,0.
panel_1
! Load cases
2
-1.,-1.,0.,0.,0.
1,-6.881E-3,-6.881E-3,-6.881E-3,-6.881E-3
0.,0.,0.,0.,1.
0
1,376.5,184
N
```

ej_pansa_1.RES

26-03-2008, Pansa 5.0

PANEL : panel_1

SANDWICH PANEL PROPERTIES

LAYUP

NUM LAY	LAY THICKN.	M. PLANE DISTAN.	(Mean Values)			POISSON RATIO	LAY ORIE.	ALLOW. TENSION	MICROSTRAINS COMPRESSION	(B Values)		
			E1 (GxZ)	E2 (GyZ)	G (Ecc)					ALLOWABLE	MODULUS	
										E1	E2	G
1	0.125	5.688	122000.	10000.	4700.	0.30	45	12500.	9500.	109000.	9000.	4500.
2	0.125	5.562	122000.	10000.	4700.	0.30	-45	12500.	9500.	109000.	9000.	4500.
3	0.125	5.438	122000.	10000.	4700.	0.30	0	12500.	9500.	109000.	9000.	4500.
4	0.125	5.312	122000.	10000.	4700.	0.30	90	12500.	9500.	109000.	9000.	4500.
5	0.125	5.188	122000.	10000.	4700.	0.30	-45	12500.	9500.	109000.	9000.	4500.
6	0.125	5.062	122000.	10000.	4700.	0.30	45	12500.	9500.	109000.	9000.	4500.
7	10.0000	0.0000	0.4820E+02	0.2410E+02	0.1930E+03							
8	0.125	-5.062	122000.	10000.	4700.	0.30	45	12500.	9500.	109000.	9000.	4500.
9	0.125	-5.188	122000.	10000.	4700.	0.30	-45	12500.	9500.	109000.	9000.	4500.
10	0.125	-5.312	122000.	10000.	4700.	0.30	90	12500.	9500.	109000.	9000.	4500.
11	0.125	-5.438	122000.	10000.	4700.	0.30	0	12500.	9500.	109000.	9000.	4500.
12	0.125	-5.562	122000.	10000.	4700.	0.30	-45	12500.	9500.	109000.	9000.	4500.
13	0.125	-5.688	122000.	10000.	4700.	0.30	45	12500.	9500.	109000.	9000.	4500.

INPLANE STIFFNESS MATRIX (Mean Values)

UPPER FACE

LOWER FACE

36350.8	15783.8	0.0	36350.8	15783.8	0.0
15783.8	36350.8	0.0	15783.8	36350.8	0.0
0.0	0.0	17042.1	0.0	0.0	17042.1

INPLANE ELASTIC MODULUS (Mean Values)

	Ex	Ey	Exy	Nux	Nuy
UP&LOW FACES:	38401.92	38401.97	22722.74	0.43	0.43

SANDWICH STIFFNESS MATRIX (Mean Values)

INPLANE STIFFNESS MATRIX

COUPLING MATRIX

BENDING STIFFNESS MATRIX

72711.6	31567.5	0.0	0.0	0.0	0.0	2122275.5	914049.0	881.5
31567.5	72701.7	0.0	0.0	0.0	0.0	914049.0	2084289.5	881.0
0.0	0.0	34084.1	0.0	0.0	0.0	881.5	881.0	986871.9
Ex	Ey	Exy	Nux	Nuy		Dx	Dy	Dxy
5130.9	5130.2	2963.8	0.43	0.43		1721426.4	1690615.0	986871.9

SANDWICH STIFFNESS MATRIX (B Values)

INPLANE STIFFNESS MATRIX

COUPLING MATRIX

BENDING STIFFNESS MATRIX

65311.8	27941.1	0.0	0.0	0.0	0.0	1906179.5	809041.9	787.1
27941.1	65301.9	0.0	0.0	0.0	0.0	809041.9	1872253.0	786.6
0.0	0.0	30610.8	0.0	0.0	0.0	787.1	786.6	886295.8
Ex	Ey	Exy				Dx	Dy	Dxy
4639.7	4639.0	2661.8				1556574.6	1528870.5	886295.8

SANDWICH PANEL GEOMETRY

DIMENSION X	:	753.000
DIMENSION Y	:	368.000
UPPER FACE THICKNESS	:	0.750
CORE THICKNESS	:	10.000
LOWER FACE THICKNESS	:	0.750
TOTAL THICKNESS	:	11.500
CELL SIZE	:	4.800

BOUNDARY CONDITIONS : SIMPLY SUPPORTED

NO GENERALLY ORTOTROPIC PANEL

PANEL : panel_1

LOAD CASE : 1

INPLANE APPLIED FLOWS : FX = -1.0 FY = 0.0 FXY = 0.0

APPLIED PRESSURE LOAD : P (x= 0.0, y= 0.0) =-0.0069 P (x= 753.0, y= 0.0) =-0.0069
P (x= 753.0, y= 368.0) =-0.0069 P (x= 0.0, y= 368.0) =-0.0069

POINT : 1 WITH COORDINATES x= 376.5, y= 184.0

INTERNAL LOADS : Nx = -1.0 Ny = 0.0 Nxy = 0.0 Mx = -50.6 My = -85.6 Mxy = 0.0 Qx = 2.29E-08 Qy =-5.51E-08

OUT-OF-PLANE DISPLACEMENT = -0.899

MIDDLE-PLANE DEFORMATIONS : (-1.695E-05 , 7.359E-06 , 0.000E+00)

LAYER	MICROSTRAINS IN FIB & MATRIX DIRECTIONS			MICROSTRAINS IN ELEMENT COORD SYSTEM		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY
1	-137.422	-137.454	-154.107	-60.384	-214.491	0.032
2	-133.115	-133.083	147.587	-59.306	-206.892	0.031
3	-58.227	-199.294	0.031	-58.227	-199.294	0.031
4	-191.695	-57.148	-0.030	-57.148	-191.695	0.030
5	-120.097	-120.068	128.027	-56.069	-184.096	0.029
6	-115.729	-115.758	-121.507	-54.990	-176.497	0.029
7	0.000	0.000	0.000	0.000	0.000	0.000
8	106.141	106.169	170.120	21.095	191.215	-0.029
9	110.508	110.479	-176.640	22.174	198.814	-0.029
10	206.412	23.252	0.030	23.252	206.412	-0.030
11	24.331	214.011	-0.031	24.331	214.011	-0.031
12	123.526	123.494	-196.200	25.410	221.610	-0.031
13	127.833	127.865	202.720	26.489	229.209	-0.032

LAYER	STRESSES IN FIBRE AND MATRIX DIRECTIONS			STRESSES IN ELEMENT COORDINATE SYSTEM			PRINCIPAL STRESSES		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY	MAJOR	MINOR	SHEAR
1	-1.73055E+01	-1.80008E+00	-7.24301E-01	-8.82847E+00	-1.02771E+01	-7.75269E+00	-1.76632E+00	-1.73392E+01	7.78645E+00
2	-1.67629E+01	-1.74303E+00	6.93657E-01	-8.55931E+00	-9.94662E+00	7.50993E+00	-1.71107E+00	-1.67949E+01	7.54190E+00
3	-7.75878E+00	-2.18372E+00	1.44409E-04	-7.75878E+00	-2.18372E+00	1.44409E-04	-2.18373E+00	-7.75878E+00	2.78753E+00
4	-2.37333E+01	-1.15508E+00	-1.40979E-04	-1.15508E+00	-2.37333E+01	1.40979E-04	-1.15508E+00	-2.37333E+01	1.12891E+01
5	-1.51236E+01	-1.57257E+00	6.01727E-01	-7.74637E+00	-8.94982E+00	6.77553E+00	-1.54590E+00	-1.51503E+01	6.80219E+00
6	-1.45738E+01	-1.51595E+00	-5.71083E-01	-7.47378E+00	-8.61595E+00	-6.52891E+00	-1.49102E+00	-1.45987E+01	6.55384E+00
7	2.12832E-09	-5.12313E-09	0.00000E+00	2.12832E-09	-5.12313E-09	0.00000E+00	2.12832E-09	-5.12313E-09	0.00000E+00
8	1.33663E+01	1.39037E+00	7.99565E-01	6.57874E+00	8.17787E+00	5.98794E+00	1.34194E+01	1.33722E+00	6.04109E+00
9	1.39161E+01	1.44699E+00	-8.30209E-01	6.85133E+00	8.51175E+00	-6.23456E+00	1.39711E+01	1.39195E+00	6.28959E+00
10	2.54397E+01	8.58091E-01	1.40979E-04	8.58091E-01	2.54397E+01	-1.40979E-04	2.54397E+01	8.58091E-01	1.22908E+01
11	3.63727E+00	2.22955E+00	-1.44409E-04	3.63727E+00	2.22955E+00	-1.44409E-04	3.63727E+00	2.22955E+00	7.03860E-01
12	1.55544E+01	1.61745E+00	-9.22140E-01	7.66427E+00	9.50855E+00	-6.96896E+00	1.56161E+01	1.55671E+00	7.02971E+00
13	1.60979E+01	1.67450E+00	9.52783E-01	7.93344E+00	9.83900E+00	7.21172E+00	1.61606E+01	1.61183E+00	7.27439E+00

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER						POINT					WRINKLING				
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
0	90	-550	0.00E+0	-2.17E+1	0.00E+0	0	-550	0.00E+0	-2.17E+1	0.00E+0	0	-550	0.00E+0	-2.17E+1	0.00E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPLING			CONCENTRATION		
	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF	RF	RF	RF
(UP)	-415.3	533.4	39.11	-2310.8	1423.1	131.76	-160.7	15.13	17.25	17.25	20.80	
(LO)	-450.5	533.4	999.99	-2310.8	1423.1	999.99	-160.7	999.99				

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

PANEL : panel_1

LOAD CASE : 2

INPLANE APPLIED FLOWS : FX = 0.0 FY = 0.0 FXY = 1.0

POINT : 1 WITH COORDINATES x= 376.5, y= 184.0

INTERNAL LOADS : Nx = 0.0 Ny = 0.0 Nxy = 1.0 Mx = 0.0 My = 0.0 Mxy = 0.0 Qx = 0.00E+00 Qy = 0.00E+00

OUT-OF-PLANE DISPLACEMENT = 0.000

MIDDLE-PLANE DEFORMATIONS : (0.000E+00 , 0.000E+00 , 2.934E-05)

LAYER	MICROSTRAINS IN FIB & MATRIX DIRECTIONS			MICROSTRAINS IN ELEMENT COORD SYSTEM		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY
1	14.670	-14.670	0.000	0.000	0.000	29.339
2	-14.670	14.670	0.000	0.000	0.000	29.339
3	0.000	0.000	29.339	0.000	0.000	29.339
4	0.000	0.000	-29.339	0.000	0.000	29.339
5	-14.670	14.670	0.000	0.000	0.000	29.339
6	14.670	-14.670	0.000	0.000	0.000	29.339
7	0.000	0.000	0.000	0.000	0.000	0.000
8	14.670	-14.670	0.000	0.000	0.000	29.339
9	-14.670	14.670	0.000	0.000	0.000	29.339
10	0.000	0.000	-29.339	0.000	0.000	29.339
11	0.000	0.000	29.339	0.000	0.000	29.339
12	-14.670	14.670	0.000	0.000	0.000	29.339
13	14.670	-14.670	0.000	0.000	0.000	29.339

LAYER	STRESSES IN FIBRE AND MATRIX DIRECTIONS			STRESSES IN ELEMENT COORDINATE SYSTEM			PRINCIPAL STRESSES		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY	MAJOR	MINOR	SHEAR
1	1.75866E+00	-1.03450E-01	0.00000E+00	8.27603E-01	8.27603E-01	9.31053E-01	1.75866E+00	-1.03450E-01	9.31053E-01
2	-1.75866E+00	1.03450E-01	0.00000E+00	-8.27603E-01	-8.27603E-01	9.31053E-01	1.03450E-01	-1.75866E+00	9.31053E-01
3	0.00000E+00	0.00000E+00	1.37894E-01	0.00000E+00	0.00000E+00	1.37894E-01	1.37894E-01	-1.37894E-01	1.37894E-01
4	0.00000E+00	0.00000E+00	-1.37894E-01	0.00000E+00	0.00000E+00	1.37894E-01	1.37894E-01	-1.37894E-01	1.37894E-01
5	-1.75866E+00	1.03450E-01	0.00000E+00	-8.27603E-01	-8.27603E-01	9.31053E-01	1.03450E-01	-1.75866E+00	9.31053E-01
6	1.75866E+00	-1.03450E-01	0.00000E+00	8.27603E-01	8.27603E-01	9.31053E-01	1.75866E+00	-1.03450E-01	9.31053E-01
7	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
8	1.75866E+00	-1.03450E-01	0.00000E+00	8.27603E-01	8.27603E-01	9.31053E-01	1.75866E+00	-1.03450E-01	9.31053E-01
9	-1.75866E+00	1.03450E-01	0.00000E+00	-8.27603E-01	-8.27603E-01	9.31053E-01	1.03450E-01	-1.75866E+00	9.31053E-01
10	0.00000E+00	0.00000E+00	-1.37894E-01	0.00000E+00	0.00000E+00	1.37894E-01	1.37894E-01	-1.37894E-01	1.37894E-01
11	0.00000E+00	0.00000E+00	1.37894E-01	0.00000E+00	0.00000E+00	1.37894E-01	1.37894E-01	-1.37894E-01	1.37894E-01
12	-1.75866E+00	1.03450E-01	0.00000E+00	-8.27603E-01	-8.27603E-01	9.31053E-01	1.03450E-01	-1.75866E+00	9.31053E-01
13	1.75866E+00	-1.03450E-01	0.00000E+00	8.27603E-01	8.27603E-01	9.31053E-01	1.75866E+00	-1.03450E-01	9.31053E-01

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER						POINT					WRINKLING				
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
45	-45	-52	-1.45E+0	-1.45E+0	1.45E+0	45	-52	-1.45E+0	-1.45E+0	1.45E+0	45	-52	-1.45E+0	-1.45E+0	1.45E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPLING		CONCENTRATION		
	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF fiber	RF point	RF Wrin
(UP)	-450.5	533.4	800.15	-2310.8	1423.1	999.99	-160.7	999.99	179.45	179.45	209.05
(LO)	-450.5	533.4	800.15	-2310.8	1423.1	999.99	-160.7	999.99			

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

GENERAL BUCKLING ALLOWABLE LOADS

CASE	APPLIED FLOW			BUCKLING FLOW			NOX NOY		RF
	FX	FY	FX	FY	FX	FY			
1	-1.0	0.0	0.0	-337.0	0.0	0.0	3	1	336.97
2	0.0	0.0	1.0	0.0	0.0	324.9	2	2	324.95

MINIMUM GENERAL BUCKLING RF = 324.95 FOR LOAD CASE 2

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (ONLY WITH INPLANE LOADS)

CASE			FIBER				POINT				WRINKLING					
No	Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
1	90	0	-45	-1.79E+0	0.00E+0	0.00E+0	90	-45	-1.79E+0	0.00E+0	0.00E+0	90	-45	-1.79E+0	0.00E+0	0.00E+0
2	45	-45	-52	-1.45E+0	-1.45E+0	1.45E+0	45	-52	-1.45E+0	-1.45E+0	1.45E+0	45	-52	-1.45E+0	-1.45E+0	1.45E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (WHOLE PANEL ONLY WITH INPLANE LOADS)

CASE	GENERAL BUCKLING		WRINKLING			DIMPLING			SHEAR CRIMPLING		CONCENTRATION		
	No	RF	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF fiber	RF point	RF wrin
1	336.97	(UP) (LO)	-567.6 -567.6	533.4 533.4	851.61 851.61	-2310.8 -2310.8	1423.1 1423.1	999.99 999.99	-160.7 -160.7	482.08 482.08	208.24 208.24	208.24 208.24	316.36 316.36
2	324.95	(UP) (LO)	-450.5 -450.5	533.4 533.4	800.15 800.15	-2310.8 -2310.8	1423.1 1423.1	999.99 999.99	-160.7 -160.7	999.99 999.99	179.45 179.45	179.45 179.45	209.05 209.05

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

SUMMARY OF RESERVE FACTORS FOR PANEL: panel_1

LOAD CASE No 1:

RF BUCKLING: 336.97

	INPLANE		INPLANE+NORMAL LOADS			
	FACE	RF	FACE	X	Y	RF
WRINKLING:	UP&LO	851.61	UPPER	376.50	184.00	39.11
DIMPLING :	UP&LO	999.99	UPPER	376.50	184.00	131.76
CRIMPLING:	UP&LO	482.08	UPPER	376.50	184.00	15.13
CONC. Fib:	UPPER	208.24	UPPER	376.50	184.00	17.25
CONC. Pnt:	UPPER	208.24	UPPER	376.50	184.00	17.25
CONC. Wrk:	UPPER	316.36	UPPER	376.50	184.00	20.80

C.A.S.A. Criterium: YES

NO: UPPER 376.50 184.00

LOAD CASE No 2:

RF BUCKLING: 324.95

	INPLANE		INPLANE+NORMAL LOADS			
	FACE	RF	FACE	X	Y	RF
WRINKLING:	UP&LO	800.15	UP&LO	376.50	184.00	800.15
DIMPLING :	UP&LO	999.99	UP&LO	376.50	184.00	999.99
CRIMPLING:	UP&LO	999.99	UP&LO	376.50	184.00	999.99
CONC. Fib:	UPPER	179.45	UPPER	376.50	184.00	179.45
CONC. Pnt:	UPPER	179.45	UPPER	376.50	184.00	179.45
CONC. Wrk:	UPPER	209.05	UPPER	376.50	184.00	209.05

C.A.S.A. Criterium: YES

YES

ej_pansa_1.AUX

PANEL : panel_1 LOAD CASE : 1

Displacement and curvature on the panel centre and shear loads on the centre of the sides

ITER	NX	NY	Wc	Kmx	Kmy	Kmxy	Kx	Ky	Kxy	Qx	Qy
1	7	5	-0.90	-8.27E-06	-3.73E-05	4.23E-09	-1.15E-05	-5.40E-05	4.07E-09	-8.58E-01	-1.02E+00
2	7	5	-0.90	-7.58E-06	-3.77E-05	5.65E-09	-8.63E-06	-6.08E-05	5.84E-09	-9.26E-01	-1.04E+00

Auxiliar information about the buckling calculation:

Number of iterations : 2

Relative difference : 0.00

ITER	NX	NY	FACTOR
1	4	3	336.968
2	6	5	336.969

BUCKLING MODE	RESERVE FACTOR	X DIRECTION SEMIWAVES	Y DIRECTION SEMIWAVES
1	336.969	3	1
2	353.423	2	1
3	359.596	4	1
4	389.838	5	1
5	418.270	6	1

PANEL : panel_1 LOAD CASE : 2

Displacement and curvature on the panel centre and shear loads on the centre of the sides

ITER	NX	NY	Wc	Kmx	Kmy	Kmxy	Kx	Ky	Kxy	Qx	Qy
1	5	3	0.00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Auxiliar information about the buckling calculation:

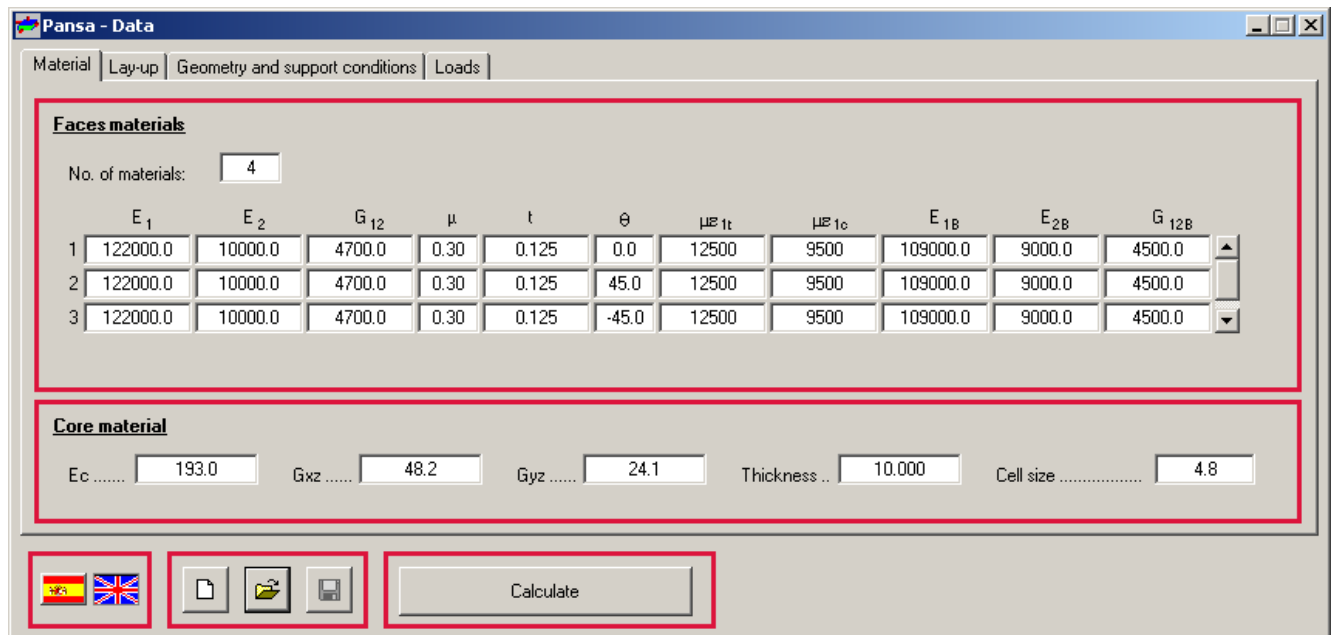
Number of iterations : 3

Relative difference : 0.48

ITER	NX	NY	FACTOR
1	4	3	341.199
2	6	5	326.511
3	8	7	324.945

BUCKLING MODE	RESERVE FACTOR	X DIRECTION SEMIWAVES	Y DIRECTION SEMIWAVES
1	324.945	2	2
2	327.784	3	3
3	359.451	4	2
4	361.116	4	4
5	387.418	5	3

If the PC version is used and the input file shown before is open, the program shows the following on the screen:



Pansa - Data

Material | Lay-up | Geometry and support conditions | Loads

Faces materials

No. of materials: 4

	E_1	E_2	G_{12}	μ	t	θ	$\mu \epsilon_{1t}$	$\mu \epsilon_{1c}$	E_{1B}	E_{2B}	G_{12B}
1	122000.0	10000.0	4700.0	0.30	0.125	0.0	12500	9500	109000.0	9000.0	4500.0
2	122000.0	10000.0	4700.0	0.30	0.125	45.0	12500	9500	109000.0	9000.0	4500.0
3	122000.0	10000.0	4700.0	0.30	0.125	-45.0	12500	9500	109000.0	9000.0	4500.0

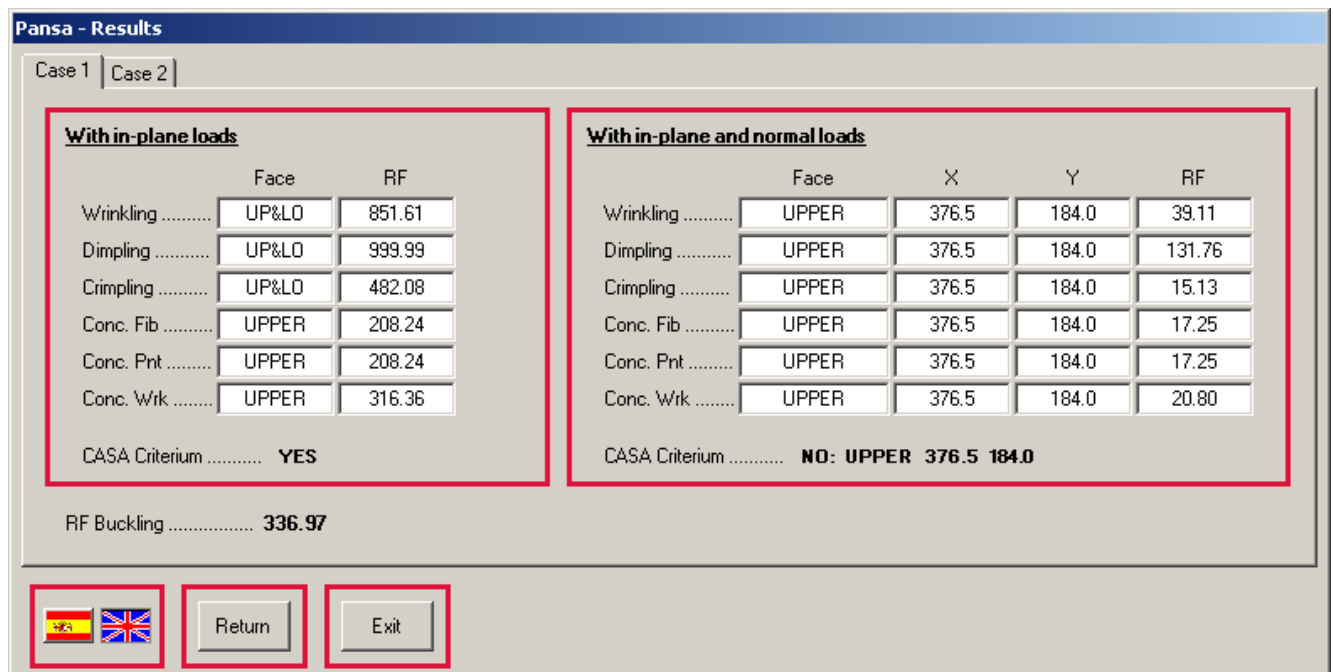
Core material

E_c 193.0 G_{xz} 48.2 G_{yz} 24.1 Thickness .. 10.000 Cell size 4.8

Spanish | English | File icons | Calculate

Figure 8.1. Example 1 input data

If the 'Calculate' button is pressed the results are shown:



Pansa - Results

Case 1 | Case 2

With in-plane loads

	Face	RF
Wrinkling	UP&LO	851.61
Dimpling	UP&LO	999.99
Crimpling	UP&LO	482.08
Conc. Fib	UPPER	208.24
Conc. Pnt	UPPER	208.24
Conc. Wrk	UPPER	316.36

CASA Criterium **YES**

RF Buckling **336.97**

With in-plane and normal loads

	Face	X	Y	RF
Wrinkling	UPPER	376.5	184.0	39.11
Dimpling	UPPER	376.5	184.0	131.76
Crimpling	UPPER	376.5	184.0	15.13
Conc. Fib	UPPER	376.5	184.0	17.25
Conc. Pnt	UPPER	376.5	184.0	17.25
Conc. Wrk	UPPER	376.5	184.0	20.80

CASA Criterium **NO: UPPER 376.5 184.0**

Spanish | English | Return | Exit

Figure 8.2. Example 1 output results

8.2 Example 2

The second panel has the same dimensions, core and lower face as the previous panel, but the upper face has been changed to an 8 layers laminate (45/-45/0/45/-45/90/-45/45), so it is not symmetrical. It is clamped and subjected to the same load cases as the first panel. In this panel the analysis is required on the centre of the panel and on the point whose coordinates are: $x = 188$ mm and $y = 184$ mm.

The data input file is:

ej_pansa_2.dat

```
! Materials
4
1,122000,10000,4700,.3,.125,0,12500E-6,9500E-6,109000,9000,4500
2,122000,10000,4700,.3,.125,45,12500E-6,9500E-6,109000,9000,4500
3,122000,10000,4700,.3,.125,-45,12500E-6,9500E-6,109000,9000,4500
4,122000,10000,4700,.3,.125,90,12500E-6,9500E-6,109000,9000,4500
! Laminate
0
8
2,3,1,2,3,4,3,2
d
6
2,3,1,4,3,2
! Core properties
193.0,48.2,24.1,10.0,4.8
! Panel dimensions
753,368
! Support conditions
4,0.,0.
panel_2
! Load cases
2
-1.,-1.,0.,0.,0.
1,-6.881E-3,-6.881E-3,-6.881E-3,-6.881E-3
0.,0.,0.,0.,1.
0
2,376.5,184,188,184
N
```

The output files are the following:

ej_pansa_2.RES

26-03-2008, Pansa 5.0

PANEL : panel_2

SANDWICH PANEL PROPERTIES

LAYUP

NUM LAY	LAY THICKN.	M. PLANE DISTAN.	(Mean Values)			POISSON RATIO	LAY ORIE.	ALLOW. TENSION	MICROSTRAINS COMPRESSION	(B Values)		
			E1 (Gxz)	E2 (Gyz)	G (Ecc)					ALLOWABLE	MODULUS	
										E1	E2	G
1	0.125	5.812	122000.	10000.	4700.	0.30	45	12500.	9500.	109000.	9000.	4500.
2	0.125	5.688	122000.	10000.	4700.	0.30	-45	12500.	9500.	109000.	9000.	4500.
3	0.125	5.562	122000.	10000.	4700.	0.30	0	12500.	9500.	109000.	9000.	4500.
4	0.125	5.438	122000.	10000.	4700.	0.30	45	12500.	9500.	109000.	9000.	4500.
5	0.125	5.312	122000.	10000.	4700.	0.30	-45	12500.	9500.	109000.	9000.	4500.
6	0.125	5.188	122000.	10000.	4700.	0.30	90	12500.	9500.	109000.	9000.	4500.
7	0.125	5.062	122000.	10000.	4700.	0.30	-45	12500.	9500.	109000.	9000.	4500.
8	0.125	4.938	122000.	10000.	4700.	0.30	45	12500.	9500.	109000.	9000.	4500.
9	10.0000	-0.1250	0.4820E+02	0.2410E+02	0.1930E+03							
10	0.125	-5.188	122000.	10000.	4700.	0.30	45	12500.	9500.	109000.	9000.	4500.
11	0.125	-5.312	122000.	10000.	4700.	0.30	-45	12500.	9500.	109000.	9000.	4500.
12	0.125	-5.438	122000.	10000.	4700.	0.30	90	12500.	9500.	109000.	9000.	4500.
13	0.125	-5.562	122000.	10000.	4700.	0.30	0	12500.	9500.	109000.	9000.	4500.
14	0.125	-5.688	122000.	10000.	4700.	0.30	-45	12500.	9500.	109000.	9000.	4500.
15	0.125	-5.812	122000.	10000.	4700.	0.30	45	12500.	9500.	109000.	9000.	4500.

INPLANE STIFFNESS MATRIX (Mean Values)

UPPER FACE			LOWER FACE		
46214.9	23297.9	0.0	36350.8	15783.8	0.0
23297.9	46215.0	0.0	15783.8	36350.8	0.0
0.0	0.0	24975.6	0.0	0.0	17042.1

INPLANE ELASTIC MODULUS (Mean Values)

	Ex	Ey	Exy	Nux	Nuy
UPPER FACE:	32398.74	32219.28	24899.73	0.50	0.50
LOWER FACE:	38401.92	38401.97	22722.74	0.43	0.43

SANDWICH STIFFNESS MATRIX (Mean Values)

INPLANE STIFFNESS MATRIX			COUPLING MATRIX			BENDING STIFFNESS MATRIX		
82575.7	39081.6	0.0	50237.4	38415.3	440.8	2477954.2	1153828.9	5839.9
39081.6	82565.8	0.0	38415.3	46712.7	440.8	1153828.9	2401623.8	5839.5
0.0	0.0	42017.6	440.8	440.8	40512.5	5839.9	5839.5	1240560.6
Ex	Ey	Exy	Nux	Nuy		Dx	Dy	Dxy
5417.7	5423.6	3463.4	0.47	0.47		1923612.1	1864357.5	1240560.6

SANDWICH STIFFNESS MATRIX (B Values)

INPLANE STIFFNESS MATRIX			COUPLING MATRIX			BENDING STIFFNESS MATRIX		
74207.0	34586.3	0.0	45303.5	33971.9	393.5	2226573.5	1021112.5	5214.5
34586.3	74197.1	0.0	33971.8	42156.4	393.6	1021112.4	2158408.0	5214.1
0.0	0.0	37700.9	393.5	393.6	36196.6	5214.5	5214.1	1113121.5
Ex	Ey	Exy	Nux	Nuy		Dx	Dy	Dxy
4909.9	4915.2	3108.4				1743499.5	1690123.1	1113121.5

SANDWICH PANEL GEOMETRY

DIMENSION X : 753.000
DIMENSION Y : 368.000
UPPER FACE THICKNESS : 1.000
CORE THICKNESS : 10.000
LOWER FACE THICKNESS : 0.750
TOTAL THICKNESS : 11.750
CELL SIZE : 4.800

PLATE EDGE SUPPORTS : CLAMPED

NO GENERALLY ORTOTROPIC PANEL

PANEL : panel_2

LOAD CASE : 1

INPLANE APPLIED FLOWS : FX = -1.0 FY = 0.0 FXY = 0.0

APPLIED PRESSURE LOAD : P (x= 0.0, y= 0.0) =-0.0069 P (x= 753.0, y= 0.0) =-0.0069
P (x= 753.0, y= 368.0) =-0.0069 P (x= 0.0, y= 368.0) =-0.0069

POINT : 1 WITH COORDINATES x= 376.5, y= 184.0

INTERNAL LOADS : Nx = -1.0 Ny = 0.0 Nxy = 0.0 Mx = -142.4 My = -281.6 Mxy = -2.9 Qx = 7.94E-10 Qy = 9.22E-09

OUT-OF-PLANE DISPLACEMENT = -0.917

MIDDLE-PLANE DEFORMATIONS : (1.629E-05 , 6.031E-05 , 3.122E-06)

LAYER	MICROSTRAINS IN FIB & MATRIX DIRECTIONS			MICROSTRAINS IN ELEMENT COORD SYSTEM		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY
1	-316.984	-309.787	-607.394	-9.689	-617.083	-7.197
2	-302.223	-309.422	593.386	-9.130	-602.515	-7.199
3	-8.570	-587.948	-7.201	-8.570	-587.948	-7.201
4	-294.298	-287.094	-565.369	-8.011	-573.381	-7.204
5	-279.530	-286.736	551.361	-7.452	-558.813	-7.206
6	-544.246	-6.893	7.209	-6.893	-544.246	-7.209
7	-264.401	-271.611	523.345	-6.334	-529.678	-7.211
8	-264.049	-256.836	-509.337	-5.774	-515.111	-7.213
9	0.000	0.000	0.000	0.000	0.000	0.000
10	359.021	345.326	625.400	39.473	664.873	13.695
11	352.890	366.583	-639.409	40.032	679.441	13.692
12	694.008	40.591	-13.690	40.591	694.008	13.690
13	41.151	708.576	13.688	41.151	708.576	13.688
14	375.584	389.269	-681.433	41.710	723.143	13.685
15	396.831	383.148	695.442	42.269	737.711	13.683

LAYER	STRESSES IN FIBRE AND MATRIX DIRECTIONS			STRESSES IN ELEMENT COORDINATE SYSTEM			PRINCIPAL STRESSES		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY	MAJOR	MINOR	SHEAR
1	-3.98957E+01	-4.07892E+00	-2.85475E+00	-1.91326E+01	-2.48421E+01	-1.79084E+01	-3.85281E+00	-4.01218E+01	1.81345E+01
2	-3.80804E+01	-4.03062E+00	2.78891E+00	-1.82666E+01	-2.38444E+01	1.70249E+01	-3.80370E+00	-3.83073E+01	1.72518E+01
3	-2.83032E+00	-5.94908E+00	-3.38468E-02	-2.83032E+00	-5.94908E+00	-3.38468E-02	-2.82995E+00	-5.94944E+00	1.55975E+00
4	-3.70389E+01	-3.78173E+00	-2.65724E+00	-1.77531E+01	-2.30675E+01	-1.66286E+01	-3.57076E+00	-3.72498E+01	1.68395E+01
5	-3.52226E+01	-3.73349E+00	2.59140E+00	-1.68867E+01	-2.20695E+01	1.57446E+01	-3.52165E+00	-3.54345E+01	1.59564E+01
6	-6.69123E+01	-1.71431E+00	3.38803E-02	-1.71431E+00	-6.69123E+01	-3.38803E-02	-1.71429E+00	-6.69123E+01	3.25990E+01
7	-3.33175E+01	-3.53540E+00	2.45972E+00	-1.59667E+01	-2.08862E+01	1.48910E+01	-3.33361E+00	-3.35193E+01	1.50928E+01
8	-3.32297E+01	-3.38548E+00	-2.39388E+00	-1.59137E+01	-2.07015E+01	-1.49221E+01	-3.19468E+00	-3.34205E+01	1.51129E+01
9	7.30167E-11	8.47412E-10	0.00000E+00	7.30167E-11	8.47412E-10	0.00000E+00	7.30167E-11	8.47412E-10	0.00000E+00
10	4.51697E+01	4.56399E+00	2.93938E+00	2.19275E+01	2.78062E+01	2.03029E+01	4.53814E+01	4.35231E+00	2.05145E+01
11	4.44805E+01	4.75961E+00	-3.00522E+00	2.16148E+01	2.76253E+01	-1.98604E+01	4.47066E+01	4.53352E+00	2.00865E+01
12	8.54209E+01	2.50643E+00	-6.43429E-02	2.50643E+00	8.54209E+01	6.43429E-02	8.54210E+01	2.50638E+00	4.14573E+01
13	7.19921E+00	7.26279E+00	6.43317E-02	7.19921E+00	7.26279E+00	6.43317E-02	7.30275E+00	7.15924E+00	7.17575E-02
14	4.73382E+01	5.05675E+00	-3.20274E+00	2.29948E+01	2.94002E+01	-2.11408E+01	4.75795E+01	4.81552E+00	2.13820E+01
15	4.99312E+01	5.05930E+00	3.26857E+00	2.42267E+01	3.07638E+01	2.24359E+01	5.01680E+01	4.82246E+00	2.26728E+01

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER						POINT					WRINKLING				
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
0	90	-1525	0.00E+0	-5.26E+1	0.00E+0	0	-1525	0.00E+0	-5.26E+1	0.00E+0	0	-1525	0.00E+0	-5.26E+1	0.00E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPING			CONCENTRATION		
	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF	RF	RF	RF
(UP)	-406.6	525.2	15.62	-3749.8	1843.4	94.88	-137.7	5.29	6.23	6.23	8.08	
(LO)	-450.5	533.4	999.99	-2310.8	1423.1	999.99	-137.7	999.99				

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

POINT : 2 WITH COORDINATES x= 188.0, y= 184.0

INTERNAL LOADS : Nx = -1.0 Ny = 0.0 Nxy = 0.0 Mx = -127.5 My = -216.9 Mxy = -6.3 Qx = 4.75E-05 Qy = -1.49E-02

OUT-OF-PLANE DISPLACEMENT = -0.685

MIDDLE-PLANE DEFORMATIONS : (1.248E-05 , 4.831E-05 , 5.691E-06)

LAYER	MICROSTRAINS IN FIB & MATRIX DIRECTIONS			MICROSTRAINS IN ELEMENT COORD SYSTEM		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY
1	-264.684	-244.388	-386.346	-61.363	-447.709	-20.295
2	-238.260	-258.557	377.268	-59.774	-437.042	-20.297
3	-58.185	-426.376	-20.299	-58.185	-426.376	-20.299
4	-246.303	-226.003	-359.113	-56.597	-415.709	-20.300
5	-219.875	-240.176	350.035	-55.008	-405.043	-20.302
6	-394.376	-53.419	20.303	-53.419	-394.376	-20.303
7	-207.618	-227.923	331.879	-51.830	-383.710	-20.305
8	-221.796	-201.489	-322.802	-50.242	-373.043	-20.306
9	0.091	-56.731	0.000	0.091	-56.731	0.000
10	300.833	268.542	412.618	78.379	490.996	32.291
11	274.670	306.960	-421.695	79.967	501.663	32.290
12	512.329	81.556	-32.288	81.556	512.329	32.288
13	83.145	522.996	32.286	83.145	522.996	32.286
14	293.056	325.340	-448.929	84.734	533.662	32.285
15	331.467	299.184	458.007	86.322	544.329	32.283

LAYER	STRESSES IN FIBRE AND MATRIX DIRECTIONS			STRESSES IN ELEMENT COORDINATE SYSTEM			PRINCIPAL STRESSES		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY	MAJOR	MINOR	SHEAR
1	-3.32700E+01	-3.26200E+00	-1.81583E+00	-1.64502E+01	-2.00818E+01	-1.50040E+01	-3.15252E+00	-3.33795E+01	1.51135E+01
2	-3.00651E+01	-3.32487E+00	1.77316E+00	-1.49218E+01	-1.84682E+01	1.33701E+01	-3.20781E+00	-3.01822E+01	1.34872E+01
3	-8.44000E+00	-4.47130E+00	-9.54034E-02	-8.44000E+00	-4.47130E+00	-9.54034E-02	-4.46901E+00	-8.44229E+00	1.98664E+00
4	-3.09553E+01	-3.02123E+00	-1.68783E+00	-1.53005E+01	-1.86761E+01	-1.39671E+01	-2.91961E+00	-3.10570E+01	1.40687E+01
5	-2.77499E+01	-3.08414E+00	1.64516E+00	-1.37719E+01	-1.70622E+01	1.23329E+01	-2.97489E+00	-2.78592E+01	1.24421E+01
6	-4.86330E+01	-1.73008E+00	9.54254E-02	-1.73008E+00	-4.86330E+01	-9.54254E-02	-1.72989E+00	-4.86331E+01	2.34516E+01
7	-2.62065E+01	-2.92365E+00	1.55983E+00	-1.30052E+01	-1.61249E+01	1.16414E+01	-2.81961E+00	-2.63105E+01	1.17454E+01
8	-2.78691E+01	-2.70020E+00	-1.51717E+00	-1.37675E+01	-1.68018E+01	-1.25845E+01	-2.60908E+00	-2.79603E+01	1.26756E+01
9	4.37042E-06	-1.36722E-03	0.00000E+00	4.37042E-06	-1.36722E-03	0.00000E+00	4.37042E-06	-1.36722E-03	0.00000E+00
10	3.77860E+01	3.61458E+00	1.93930E+00	1.87610E+01	2.26396E+01	1.70857E+01	3.78957E+01	3.50487E+00	1.71954E+01
11	3.46865E+01	3.92255E+00	-1.98197E+00	1.73226E+01	2.12865E+01	-1.53820E+01	3.48137E+01	3.79538E+00	1.55092E+01
12	6.32152E+01	2.37003E+00	-1.51753E-01	2.37003E+00	6.32152E+01	1.51753E-01	6.32156E+01	2.36965E+00	3.04230E+01
13	1.17997E+01	5.52012E+00	1.51746E-01	1.17997E+01	5.52012E+00	1.51746E-01	1.18034E+01	5.51645E+00	3.14346E+00
14	3.70018E+01	4.16328E+00	-2.10997E+00	1.84726E+01	2.26925E+01	-1.64192E+01	3.71368E+01	4.02826E+00	1.65543E+01
15	4.16438E+01	4.01587E+00	2.15263E+00	2.06772E+01	2.49824E+01	1.88139E+01	4.17665E+01	3.89312E+00	1.89367E+01

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER						POINT					WRINKLING				
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
175	90	-1125	-2.98E-1	-3.90E+1	-3.41E+0	175	-1125	-2.98E-1	-3.90E+1	-3.41E+0	175	-1125	-2.98E-1	-3.90E+1	-3.41E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPLING		CONCENTRATION		
	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF fiber	RF point	RF Wrin
(UP)	-397.2	525.2	19.81	-3749.8	1843.4	116.51	-137.7	6.87	8.44	8.44	10.85
(LO)	-450.5	533.4	727.09	-2310.8	1423.1	999.99	-137.7	999.99			

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

PANEL : panel_2

LOAD CASE : 2

INPLANE APPLIED FLOWS : FX = 0.0 FY = 0.0 FXY = 1.0

POINT : 1 WITH COORDINATES x= 376.5, y= 184.0

INTERNAL LOADS : Nx = 0.0 Ny = 0.0 Nxy = 1.0 Mx = 0.0 My = 0.0 Mxy = 1.0 Qx = 0.00E+00 Qy = 0.00E+00

OUT-OF-PLANE DISPLACEMENT = 0.000

MIDDLE-PLANE DEFORMATIONS : (0.000E+00 , 0.000E+00 , 2.380E-05)

LAYER	MICROSTRAINS IN FIB & MATRIX DIRECTIONS			MICROSTRAINS IN ELEMENT COORD SYSTEM		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY
1	11.900	-11.900	0.000	0.000	0.000	23.800
2	-11.900	11.900	0.000	0.000	0.000	23.800
3	0.000	0.000	23.800	0.000	0.000	23.800
4	11.900	-11.900	0.000	0.000	0.000	23.800
5	-11.900	11.900	0.000	0.000	0.000	23.800
6	0.000	0.000	-23.800	0.000	0.000	23.800
7	-11.900	11.900	0.000	0.000	0.000	23.800
8	11.900	-11.900	0.000	0.000	0.000	23.800
9	0.000	0.000	0.000	0.000	0.000	0.000
10	11.900	-11.900	0.000	0.000	0.000	23.800
11	-11.900	11.900	0.000	0.000	0.000	23.800
12	0.000	0.000	-23.800	0.000	0.000	23.800
13	0.000	0.000	23.800	0.000	0.000	23.800
14	-11.900	11.900	0.000	0.000	0.000	23.800
15	11.900	-11.900	0.000	0.000	0.000	23.800

LAYER	STRESSES IN FIBRE AND MATRIX DIRECTIONS			STRESSES IN ELEMENT COORDINATE SYSTEM			PRINCIPAL STRESSES		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY	MAJOR	MINOR	SHEAR
1	1.42660E+00	-8.39174E-02	0.00000E+00	6.71339E-01	6.71339E-01	7.55257E-01	1.42660E+00	-8.39174E-02	7.55257E-01
2	-1.42660E+00	8.39174E-02	0.00000E+00	-6.71339E-01	-6.71339E-01	7.55257E-01	8.39174E-02	-1.42660E+00	7.55257E-01
3	0.00000E+00	0.00000E+00	1.11858E-01	0.00000E+00	0.00000E+00	1.11858E-01	1.11858E-01	-1.11858E-01	1.11858E-01
4	1.42660E+00	-8.39174E-02	0.00000E+00	6.71339E-01	6.71339E-01	7.55257E-01	1.42660E+00	-8.39174E-02	7.55257E-01
5	-1.42660E+00	8.39174E-02	0.00000E+00	-6.71339E-01	-6.71339E-01	7.55257E-01	8.39174E-02	-1.42660E+00	7.55257E-01
6	0.00000E+00	0.00000E+00	-1.11858E-01	0.00000E+00	0.00000E+00	1.11858E-01	1.11858E-01	-1.11858E-01	1.11858E-01
7	-1.42660E+00	8.39174E-02	0.00000E+00	-6.71339E-01	-6.71339E-01	7.55257E-01	8.39174E-02	-1.42660E+00	7.55257E-01
8	1.42660E+00	-8.39174E-02	0.00000E+00	6.71339E-01	6.71339E-01	7.55257E-01	1.42660E+00	-8.39174E-02	7.55257E-01
9	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
10	1.42660E+00	-8.39174E-02	0.00000E+00	6.71339E-01	6.71339E-01	7.55257E-01	1.42660E+00	-8.39174E-02	7.55257E-01
11	-1.42660E+00	8.39174E-02	0.00000E+00	-6.71339E-01	-6.71339E-01	7.55257E-01	8.39174E-02	-1.42660E+00	7.55257E-01
12	0.00000E+00	0.00000E+00	-1.11858E-01	0.00000E+00	0.00000E+00	1.11858E-01	1.11858E-01	-1.11858E-01	1.11858E-01
13	0.00000E+00	0.00000E+00	1.11858E-01	0.00000E+00	0.00000E+00	1.11858E-01	1.11858E-01	-1.11858E-01	1.11858E-01
14	-1.42660E+00	8.39174E-02	0.00000E+00	-6.71339E-01	-6.71339E-01	7.55257E-01	8.39174E-02	-1.42660E+00	7.55257E-01
15	1.42660E+00	-8.39174E-02	0.00000E+00	6.71339E-01	6.71339E-01	7.55257E-01	1.42660E+00	-8.39174E-02	7.55257E-01

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER						POINT					WRINKLING				
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
45	-45	-47	-1.37E+0	-1.37E+0	1.37E+0	45	-47	-1.37E+0	-1.37E+0	1.37E+0	45	-47	-1.37E+0	-1.37E+0	1.37E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPLING			CONCENTRATION		
	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF	RF	RF	RF
(UP)	-424.9	525.2	883.63	-3749.8	1843.4	999.99	-137.7	999.99	202.09	202.09	214.54	
(LO)	-450.5	533.4	986.39	-2310.8	1423.1	999.99	-137.7	999.99				

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

POINT : 2 WITH COORDINATES x= 188.0, y= 184.0

INTERNAL LOADS : Nx = 0.0 Ny = 0.0 Nxy = 1.0 Mx = 0.0 My = 0.0 Mxy = 1.0 Qx = 0.00E+00 Qy = 0.00E+00

OUT-OF-PLANE DISPLACEMENT = 0.000

MIDDLE-PLANE DEFORMATIONS : (0.000E+00 , 0.000E+00 , 2.380E-05)

LAYER	MICROSTRAINS IN FIB & MATRIX DIRECTIONS			MICROSTRAINS IN ELEMENT COORD SYSTEM		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY
1	11.900	-11.900	0.000	0.000	0.000	23.800
2	-11.900	11.900	0.000	0.000	0.000	23.800
3	0.000	0.000	23.800	0.000	0.000	23.800
4	11.900	-11.900	0.000	0.000	0.000	23.800
5	-11.900	11.900	0.000	0.000	0.000	23.800
6	0.000	0.000	-23.800	0.000	0.000	23.800
7	-11.900	11.900	0.000	0.000	0.000	23.800
8	11.900	-11.900	0.000	0.000	0.000	23.800
9	0.000	0.000	0.000	0.000	0.000	0.000
10	11.900	-11.900	0.000	0.000	0.000	23.800
11	-11.900	11.900	0.000	0.000	0.000	23.800
12	0.000	0.000	-23.800	0.000	0.000	23.800
13	0.000	0.000	23.800	0.000	0.000	23.800
14	-11.900	11.900	0.000	0.000	0.000	23.800
15	11.900	-11.900	0.000	0.000	0.000	23.800

LAYER	STRESSES IN FIBRE AND MATRIX DIRECTIONS			STRESSES IN ELEMENT COORDINATE SYSTEM			PRINCIPAL STRESSES		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY	MAJOR	MINOR	SHEAR
1	1.42660E+00	-8.39174E-02	0.00000E+00	6.71339E-01	6.71339E-01	7.55257E-01	1.42660E+00	-8.39174E-02	7.55257E-01
2	-1.42660E+00	8.39174E-02	0.00000E+00	-6.71339E-01	-6.71339E-01	7.55257E-01	8.39174E-02	-1.42660E+00	7.55257E-01
3	0.00000E+00	0.00000E+00	1.11858E-01	0.00000E+00	0.00000E+00	1.11858E-01	1.11858E-01	-1.11858E-01	1.11858E-01
4	1.42660E+00	-8.39174E-02	0.00000E+00	6.71339E-01	6.71339E-01	7.55257E-01	1.42660E+00	-8.39174E-02	7.55257E-01
5	-1.42660E+00	8.39174E-02	0.00000E+00	-6.71339E-01	-6.71339E-01	7.55257E-01	8.39174E-02	-1.42660E+00	7.55257E-01
6	0.00000E+00	0.00000E+00	-1.11858E-01	0.00000E+00	0.00000E+00	1.11858E-01	1.11858E-01	-1.11858E-01	1.11858E-01
7	-1.42660E+00	8.39174E-02	0.00000E+00	-6.71339E-01	-6.71339E-01	7.55257E-01	8.39174E-02	-1.42660E+00	7.55257E-01
8	1.42660E+00	-8.39174E-02	0.00000E+00	6.71339E-01	6.71339E-01	7.55257E-01	1.42660E+00	-8.39174E-02	7.55257E-01
9	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
10	1.42660E+00	-8.39174E-02	0.00000E+00	6.71339E-01	6.71339E-01	7.55257E-01	1.42660E+00	-8.39174E-02	7.55257E-01
11	-1.42660E+00	8.39174E-02	0.00000E+00	-6.71339E-01	-6.71339E-01	7.55257E-01	8.39174E-02	-1.42660E+00	7.55257E-01
12	0.00000E+00	0.00000E+00	-1.11858E-01	0.00000E+00	0.00000E+00	1.11858E-01	1.11858E-01	-1.11858E-01	1.11858E-01
13	0.00000E+00	0.00000E+00	1.11858E-01	0.00000E+00	0.00000E+00	1.11858E-01	1.11858E-01	-1.11858E-01	1.11858E-01
14	-1.42660E+00	8.39174E-02	0.00000E+00	-6.71339E-01	-6.71339E-01	7.55257E-01	8.39174E-02	-1.42660E+00	7.55257E-01
15	1.42660E+00	-8.39174E-02	0.00000E+00	6.71339E-01	6.71339E-01	7.55257E-01	1.42660E+00	-8.39174E-02	7.55257E-01

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER						POINT					WRINKLING				
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
45	-45	-47	-1.37E+0	-1.37E+0	1.37E+0	45	-47	-1.37E+0	-1.37E+0	1.37E+0	45	-47	-1.37E+0	-1.37E+0	1.37E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPLING		CONCENTRATION		
	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF fiber	RF point	RF Wrin
(UP)	-424.9	525.2	883.63	-3749.8	1843.4	999.99	-137.7	999.99	202.09	202.09	214.54
(LO)	-450.5	533.4	986.39	-2310.8	1423.1	999.99	-137.7	999.99			

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

GENERAL BUCKLING ALLOWABLE LOADS

CASE	APPLIED FLOW			BUCKLING FLOW			NOX NOY		RF
	FX	FY	FX	FY	FX	FY			
1	-1.0	0.0	0.0	-405.1	0.0	0.0	3	1	405.13
2	0.0	0.0	1.0	0.0	0.0	357.9	4	2	357.91

MINIMUM GENERAL BUCKLING RF = 357.91 FOR LOAD CASE 2

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (ONLY WITH INPLANE LOADS)

CASE			FIBER				POINT				WRINKLING					
No	Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
1	90	0	-39	-1.37E+0	0.00E+0	0.00E+0	90	-39	-1.37E+0	0.00E+0	0.00E+0	90	-39	-1.37E+0	0.00E+0	0.00E+0
2	45	-45	-47	-1.37E+0	-1.37E+0	1.37E+0	45	-47	-1.37E+0	-1.37E+0	1.37E+0	45	-47	-1.37E+0	-1.37E+0	1.37E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (WHOLE PANEL ONLY WITH INPLANE LOADS)

CASE	GENERAL BUCKLING		WRINKLING			DIMPLING			SHEAR CRIMPLING		CONCENTRATION		
	No	RF	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF fiber	RF point	RF wrin
1	405.13	(UP) (LO)	-536.4 -567.6	525.2 533.4	976.73 944.59	-3749.8 -2310.8	1843.4 1423.1	999.99 999.99	-137.7 -275.4	501.56 458.33	238.56	238.56	390.75
2	357.91	(UP) (LO)	-424.9 -450.5	525.2 533.4	883.63 986.39	-3749.8 -2310.8	1843.4 1423.1	999.99 999.99	-137.7 -137.7	999.99 999.99	202.09	202.09	214.54

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

SUMMARY OF RESERVE FACTORS FOR PANEL: panel_2

LOAD CASE No 1:

RF BUCKLING: 405.13

	INPLANE		INPLANE+NORMAL LOADS			
	FACE	RF	FACE	X	Y	RF
WRINKLING:	LOWER	944.59	UPPER	376.50	184.00	15.62
DIMPLING :	UP&LO	999.99	UPPER	376.50	184.00	94.88
CRIMPLING:	LOWER	458.33	UPPER	376.50	184.00	5.29
CONC. Fib:	UPPER	238.56	UPPER	376.50	184.00	6.23
CONC. Pnt:	UPPER	238.56	UPPER	376.50	184.00	6.23
CONC. Wrk:	UPPER	390.75	UPPER	376.50	184.00	8.08

C.A.S.A. Criterium: YES

NO: UPPER 376.50 184.00
UPPER 188.00 184.00

LOAD CASE No 2:

RF BUCKLING: 357.91

	INPLANE		INPLANE+NORMAL LOADS			
	FACE	RF	FACE	X	Y	RF
WRINKLING:	UPPER	883.63	UPPER	188.00	184.00	883.63
DIMPLING :	UP&LO	999.99	UP&LO	188.00	184.00	999.99
CRIMPLING:	UP&LO	999.99	UP&LO	188.00	184.00	999.99
CONC. Fib:	UPPER	202.09	UPPER	188.00	184.00	202.09
CONC. Pnt:	UPPER	202.09	UPPER	188.00	184.00	202.09
CONC. Wrk:	UPPER	214.54	UPPER	188.00	184.00	214.54

C.A.S.A. Criterium: YES

YES

ej_pansa_2.AUX

PANEL : panel_2 LOAD CASE : 1

Displacement and curvature on the panel centre and shear loads on the centre of the sides

ITER	NX	NY	Wc	Kmx	Kmy	Kmxy	Kx	Ky	Kxy	Qx	Qy
1	7	5	-0.82	-6.03E-06	-1.20E-04	-4.69E-06	-6.03E-06	-1.20E-04	-2.24E-08	0.00E+00	0.00E+00
2	9	7	-0.85	-1.96E-05	-2.79E-05	-3.60E-06	-1.95E-05	-2.80E-05	2.20E-08	0.00E+00	0.00E+00
3	9	7	-0.92	-4.47E-06	-1.17E-04	-1.92E-06	-4.47E-06	-1.17E-04	1.90E-08	0.00E+00	0.00E+00

Auxiliar information about the buckling calculation:

Number of iterations : 5

Relative difference : 0.72

ITER	NX	NY	FACTOR
1	4	3	669.609
2	6	5	429.766
3	8	7	413.321
4	10	9	408.032
5	12	11	405.129

BUCKLING MODE	RESERVE FACTOR	X DIRECTION SEMIWAVES	Y DIRECTION SEMIWAVES
1	405.129	3	1
2	407.605	4	2
3	449.231	4	7
4	452.040	6	2
5	486.818	7	1

PANEL : panel_2 LOAD CASE : 2

Displacement and curvature on the panel centre and shear loads on the centre of the sides

ITER	NX	NY	Wc	Kmx	Kmy	Kmxy	Kx	Ky	Kxy	Qx	Qy
1	5	3	0.00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Auxiliar information about the buckling calculation:

Number of iterations : 5

Relative difference : 0.59

ITER	NX	NY	FACTOR
1	4	3	1206.882
2	6	5	391.543
3	8	7	364.362
4	10	9	360.010
5	12	11	357.906

BUCKLING MODE	RESERVE FACTOR	X DIRECTION SEMIWAVES	Y DIRECTION SEMIWAVES
1	357.906	4	2
2	360.029	3	3
3	383.326	4	4
4	383.363	6	4
5	396.241	5	3

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

Pansa - Data

Material Lay-up Geometry and support conditions Loads

To define the laminate

☒ using the stacking sequence ☐ using stiffness matrix ☐ using elastic moduli

☐ Symmetrical

Upper face

No. of plies: 8 Thickness: 1.000

Ply	1	2	3	4	5	6	7
Material	2	3	1	2	3	4	3

Lower face

No. of plies: 6 Thickness: 0.750

Ply	1	2	3	4	5	6
Material	2	3	1	4	3	2

Spanish English File Open Save Calculate

Figure 8.3. Example 2 input data

If the 'Calculate' button is pressed the results are shown:

Pansa - Results

Case 1 Case 2

With in-plane loads

	Face	RF
Wrinkling	UPPER	883.63
Dimpling	UP&LO	999.99
Crimpling	UP&LO	999.99
Conc. Fib	UPPER	202.09
Conc. Pnt	UPPER	202.09
Conc. Wrk	UPPER	214.54

CASA Criterium YES

With in-plane and normal loads

	Face	X	Y	RF
Wrinkling	UPPER	188.0	184.0	883.63
Dimpling	UP&LO	188.0	184.0	999.99
Crimpling	UP&LO	188.0	184.0	999.99
Conc. Fib	UPPER	188.0	184.0	202.09
Conc. Pnt	UPPER	188.0	184.0	202.09
Conc. Wrk	UPPER	188.0	184.0	214.54

CASA Criterium YES

RF Buckling 357.91

Spanish English Return Exit

Figure 8.4. Example 2 output results

8.3 Example 3

The third panel is like the first one but its supports have a rotation stiffness of $1.0E+4 \text{ Nmm}^2$ on the edges parallel to the x -axis and $1.0E+5 \text{ Nmm}^2$ on the edges parallel to the y -axis. It is subjected to a combination of an uniform compression flow on the x direction, a positive shear flow, both of 1 N/mm , and two normal loads of 100 N on points with coordinates $x = 188 \text{ mm}$, $y = 184 \text{ mm}$ and $x = 565 \text{ mm}$, $y = 184 \text{ mm}$. The user wants to know the vertical displacement and the internal loads on the centre point of the panel, as well as those points where the maximum vertical displacement takes place and the maximum internal moments.

The data input file is:

ej_pansa_3.dat

```
! Materials
4
1,122000,10000,4700,.3,.125,0,12500E-6,9500E-6,109000,9000,4500
2,122000,10000,4700,.3,.125,45,12500E-6,9500E-6,109000,9000,4500
3,122000,10000,4700,.3,.125,-45,12500E-6,9500E-6,109000,9000,4500
4,122000,10000,4700,.3,.125,90,12500E-6,9500E-6,109000,9000,4500
! Laminate
0
6
2,3,1,4,3,2
s
! Core properties
193.0,48.2,24.1,10.0,4.8
! Panel dimensions
753,368
! Support conditions
1,1.E+4,1.E+5
panel_3
! Load cases
1
-1.,-1.,0.,0.,1.
2, 2,100,188,184,100,565,184
1,376.5,184
si
```

The output files are the following:

NO GENERALLY ORTOTROPIC PANEL

PANEL : panel_3

LOAD CASE : 1

INPLANE APPLIED FLOWS : FX = -1.0 FY = 0.0 FXY = 1.0

APPLIED CONCENTRATED LOAD : F (x= 188.0, y= 184.0) = 100.000
F (x= 565.0, y= 184.0) = 100.000

POINT : 1 WITH COORDINATES x= 376.5, y= 184.0

INTERNAL LOADS : Nx = -1.0 Ny = 0.0 Nxy = 1.0 Mx = 2.7 My = 13.7 Mxy = 0.0 Qx = 4.35E-08 Qy = 1.07E-08

OUT-OF-PLANE DISPLACEMENT = 0.137

MIDDLE-PLANE DEFORMATIONS : (-1.695E-05 , 7.359E-06 , 2.934E-05)

LAYER	MICROSTRAINS IN FIBRE & MATRIX DIRECTIONS	MICROSTRAINS IN ELEMENT COORD SYSTEM
	NORMAL-1 NORMAL-2 SHEAR-12	NORMAL-X NORMAL-Y SHEAR-XY
1	25.814 -3.556 78.629	-28.186 50.444 29.371
2	-4.028 25.341 -76.897	-27.792 49.105 29.369
3	-27.399 47.766 29.367	-27.399 47.766 29.367
4	46.428 -27.005 -29.365	-27.005 46.428 29.365
5	-5.443 23.920 -71.701	-26.612 45.089 29.363
6	23.446 -5.914 69.969	-26.218 43.751 29.361
7	0.000 0.000 0.000	0.000 0.000 0.000
8	-3.696 -33.014 -21.356	-7.677 -29.033 29.318
9	-33.485 -4.170 23.088	-7.284 -30.371 29.316
10	-31.710 -6.890 -29.314	-31.710 -6.890 29.314
11	-6.497 -33.049 29.312	-6.497 -33.049 29.312
12	-34.900 -5.590 28.284	-6.103 -34.387 29.310
13	-6.064 -35.372 -30.016	-5.710 -35.726 29.308

LAYER	STRESSES IN FIBRE AND MATRIX DIRECTIONS	STRESSES IN ELEMENT COORDINATE SYSTEM	PRINCIPAL STRESSES
	NORMAL-1 NORMAL-2 SHEAR-12	NORMAL-X NORMAL-Y SHEAR-XY	MAJOR MINOR SHEAR
1	3.16200E+00 4.21903E-02 3.69557E-01	1.23254E+00 1.97165E+00 1.55991E+00	3.20518E+00 -9.88126E-04 1.60308E+00
2	-4.18474E-01 2.43117E-01 -3.61417E-01	2.73738E-01 3.30796E-01 4.02268E-01	-5.77626E-01 4.89947E-01 1.81602E+00
3	-3.22312E+00 3.98407E-01 1.38023E-01	3.98407E-01 1.38023E-01 4.03659E-01	-3.22837E+00 1.81602E+00 2.88151E+00
4	5.62466E+00 -1.31741E-01 -1.38014E-01	5.62466E+00 1.38014E-01 5.62797E+00	-1.35048E-01 2.88151E+00 5.31174E-01
5	-5.96643E-01 2.24529E-01 -3.36994E-01	1.50937E-01 4.10586E-01 3.45117E-01	-7.17231E-01 2.88151E+00 1.46371E+00
6	2.86385E+00 1.12807E-02 3.28853E-01	1.76642E+00 1.42629E+00 4.05053E-09	2.90127E+00 2.61396E-02 0.00000E+00
7	4.05053E-09 9.93283E-10 0.00000E+00	9.93283E-10 0.00000E+00 4.05053E-09	9.93283E-10 0.00000E+00 1.45367E-01
8	-5.54069E-01 -3.43764E-01 -1.00371E-01	-3.48545E-01 -5.49288E-01 -1.05152E-01	-3.03550E-01 -5.94283E-01 1.45367E-01
9	-4.12819E+00 -1.43211E-01 1.08512E-01	-2.02719E+00 -2.24421E+00 1.99249E+00	-1.40258E-01 -4.13115E+00 1.99544E+00
10	-3.91821E+00 -1.65253E-01 -1.37774E-01	-1.65253E-01 -3.91821E+00 1.37774E-01	-1.60202E-01 -3.92326E+00 1.88153E+00
11	-8.98389E-01 -3.52578E-01 1.37765E-01	-8.98389E-01 -3.52578E-01 1.37765E-01	-3.19777E-01 -9.31190E-01 3.05707E-01
12	-4.30636E+00 -1.61799E-01 1.32934E-01	-2.36701E+00 -2.07228E+00 2.07228E+00	-1.57539E-01 -4.31062E+00 2.07654E+00
13	-8.52215E-01 -3.74673E-01 -1.41075E-01	-4.72369E-01 -7.54519E-01 -2.38771E-01	-3.36111E-01 -8.90777E-01 2.77333E-01

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER			POINT				WRINKLING			
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
170	90	135	1.68E-1	5.41E+0	9.53E-1	165	138	3.92E-1	5.47E+0	1.46E+0
75	-79	-3.14E+0	-2.26E-1	8.42E-1						

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

WRINKLING			DIMPLING			SHEAR CRIMPLING			CONCENTRATION		
Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF	RF	RF	RF
(UP)	-567.6	533.4	634.54	-2310.8	1423.1	999.99	-321.3	970.91	92.20	90.35	167.91
(LO)	-423.3	533.4	228.97	-2310.8	1423.1	750.34	-160.7	94.64			

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

POINT OF MAXIMUM : w Mx y My. COORDINATES x= 376.5, y= 184.0

INTERNAL LOADS : Nx = -1.0 Ny = 0.0 Nxy = 1.0 Mx = 2.7 My = 13.7 Mxy = 0.0 Qx = 4.35E-08 Qy = 1.07E-08

OUT-OF-PLANE DISPLACEMENT = 0.137

MIDDLE-PLANE DEFORMATIONS : (-1.695E-05 , 7.359E-06 , 2.934E-05)

LAYER	MICROSTRAINS IN FIB & MATRIX DIRECTIONS			MICROSTRAINS IN ELEMENT COORD SYSTEM		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY
1	25.814	-3.556	78.629	-28.186	50.444	29.371
2	-4.028	25.341	-76.897	-27.792	49.105	29.369
3	-27.399	47.766	29.367	-27.399	47.766	29.367
4	46.428	-27.005	-29.365	-27.005	46.428	29.365
5	-5.443	23.920	-71.701	-26.612	45.089	29.363
6	23.446	-5.914	69.969	-26.218	43.751	29.361
7	0.000	0.000	0.000	0.000	0.000	0.000
8	-3.696	-33.014	-21.356	-7.677	-29.033	29.318
9	-33.485	-4.170	23.088	-7.284	-30.371	29.316
10	-31.710	-6.890	-29.314	-6.890	-31.710	29.314
11	-6.497	-33.049	29.312	-6.497	-33.049	29.312
12	-34.900	-5.590	28.284	-6.103	-34.387	29.310
13	-6.064	-35.372	-30.016	-5.710	-35.726	29.308

LAYER	STRESSES IN FIBRE AND MATRIX DIRECTIONS			STRESSES IN ELEMENT COORDINATE SYSTEM			PRINCIPAL STRESSES		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY	MAJOR	MINOR	SHEAR
1	3.16200E+00	4.21903E-02	3.69557E-01	1.23254E+00	1.97165E+00	1.55991E+00	3.20518E+00	-9.88126E-04	1.60308E+00
2	-4.18474E-01	2.43117E-01	-3.61417E-01	-4.49095E-01	2.73738E-01	3.30796E-01	4.02268E-01	-5.77626E-01	4.89947E-01
3	-3.22312E+00	3.98407E-01	1.38023E-01	-3.22312E+00	3.98407E-01	1.38023E-01	4.03659E-01	-3.22837E+00	1.81602E+00
4	5.62466E+00	-1.31741E-01	-1.38014E-01	-1.31741E-01	5.62466E+00	1.38014E-01	5.62797E+00	-1.35048E-01	2.88151E+00
5	-5.96643E-01	2.24529E-01	-3.36994E-01	-5.23051E-01	1.50937E-01	4.10586E-01	3.45117E-01	-7.17231E-01	5.31174E-01
6	2.86385E+00	1.12807E-02	3.28853E-01	1.10871E+00	1.76642E+00	1.42629E+00	2.90127E+00	-2.61396E-02	1.46371E+00
7	4.05053E-09	9.93283E-10	0.00000E+00	4.05053E-09	9.93283E-10	0.00000E+00	4.05053E-09	9.93283E-10	0.00000E+00
8	-5.54069E-01	-3.43764E-01	-1.00371E-01	-3.48545E-01	-5.49288E-01	-1.05152E-01	-3.03550E-01	-5.94283E-01	1.45367E-01
9	-4.12819E+00	-1.43211E-01	1.08512E-01	-2.02719E+00	-2.24421E+00	1.99249E+00	-1.40258E-01	-4.13115E+00	1.99544E+00
10	-3.91821E+00	-1.65253E-01	-1.37774E-01	-1.65253E-01	-3.91821E+00	1.37774E-01	-1.60202E-01	-3.92326E+00	1.88153E+00
11	-8.98389E-01	-3.52578E-01	1.37765E-01	-8.98389E-01	-3.52578E-01	1.37765E-01	-3.19777E-01	-9.31190E-01	3.05707E-01
12	-4.30636E+00	-1.61799E-01	1.32934E-01	-2.10115E+00	-2.36701E+00	2.07228E+00	-1.57539E-01	-4.31062E+00	2.07654E+00
13	-8.52215E-01	-3.74673E-01	-1.41075E-01	-4.72369E-01	-7.54519E-01	-2.38771E-01	-3.36111E-01	-8.90777E-01	2.77333E-01

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER				POINT				WRINKLING			
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	
170	90	135	1.68E-1	5.41E+0	9.53E-1	165	138	3.92E-1	5.47E+0	1.46E+0	75
											-79
											-3.14E+0
											-2.26E-1
											8.42E-1

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPLING		CONCENTRATION		
	F _{cw}	F _{sw}	RF	F _{cd}	F _{sd}	RF	F _{sc}	RF	RF	RF	RF
									fiber	point	Wrin
(UP)	-567.6	533.4	634.54	-2310.8	1423.1	999.99	-321.3	970.91	92.20	90.35	167.91
(LO)	-423.3	533.4	228.97	-2310.8	1423.1	750.34	-160.7	94.64			

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

POINT OF MAXIMUM Mxy. COORDINATES x= 0.0, y= 0.0

INTERNAL LOADS : Nx = -1.0 Ny = 0.0 Nxy = 1.0 Mx = 0.0 My = 0.0 Mxy = -9.5 Qx = 0.00E+00 Qy = 0.00E+00

OUT-OF-PLANE DISPLACEMENT = 0.000

MIDDLE-PLANE DEFORMATIONS : (-1.695E-05 , 7.359E-06 , 2.934E-05)

LAYER	MICROSTRAINS IN FIB & MATRIX DIRECTIONS			MICROSTRAINS IN ELEMENT COORD SYSTEM		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY
1	-17.846	8.257	24.307	-16.948	7.359	-26.103
2	7.524	-17.113	-24.307	-16.948	7.359	-24.637
3	-16.948	7.359	-23.170	-16.948	7.359	-23.170
4	7.359	-16.948	21.704	-16.948	7.359	-21.704
5	5.324	-14.913	-24.307	-16.948	7.359	-20.237
6	-14.180	4.591	24.307	-16.948	7.359	-18.771
7	0.000	0.000	0.000	0.000	0.000	0.000
8	33.930	-43.519	24.307	-16.948	7.359	77.449
9	-44.252	34.663	-24.307	-16.948	7.359	78.916
10	7.359	-16.948	-80.382	-16.948	7.359	80.382
11	-16.948	7.359	81.848	-16.948	7.359	81.848
12	-46.452	36.863	-24.307	-16.948	7.359	83.315
13	37.596	-47.185	24.307	-16.948	7.359	84.781

LAYER	STRESSES IN FIBRE AND MATRIX DIRECTIONS			STRESSES IN ELEMENT COORDINATE SYSTEM			PRINCIPAL STRESSES		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY	MAJOR	MINOR	SHEAR
1	-2.16843E+00	2.92481E-02	1.14241E-01	-1.18383E+00	-9.55351E-01	-1.09884E+00	3.51707E-02	-2.17436E+00	1.10476E+00
2	8.73006E-01	-1.49660E-01	-1.14241E-01	2.47432E-01	4.75914E-01	-5.11333E-01	8.85612E-01	-1.62266E-01	5.23939E-01
3	-2.06075E+00	2.29141E-02	-1.08899E-01	-2.06075E+00	2.29141E-02	-1.08899E-01	2.85901E-02	-2.06643E+00	1.04751E+00
4	8.53229E-01	-1.48497E-01	1.02007E-01	-1.48497E-01	8.53229E-01	-1.02007E-01	8.63511E-01	-1.58779E-01	5.11145E-01
5	6.09301E-01	-1.34148E-01	-1.14241E-01	1.23335E-01	3.51817E-01	-3.71724E-01	6.26459E-01	-1.51307E-01	3.8883E-01
6	-1.72892E+00	3.39466E-03	1.14241E-01	-9.77006E-01	-7.48524E-01	-8.66160E-01	1.08960E-02	-1.73643E+00	8.73661E-01
7	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
8	4.03871E+00	-3.35878E-01	1.14241E-01	1.73717E+00	1.96566E+00	2.18729E+00	4.04169E+00	-3.38860E-01	2.19028E+00
9	-5.33414E+00	2.15466E-01	-1.14241E-01	-2.67358E+00	-2.44509E+00	2.77480E+00	2.17817E-01	-5.33649E+00	2.77715E+00
10	8.53229E-01	-1.48497E-01	-3.77796E-01	-1.48497E-01	8.53229E-01	3.77796E-01	9.79736E-01	-2.75004E-01	6.27370E-01
11	-2.06075E+00	2.29141E-02	3.84688E-01	-2.06075E+00	2.29141E-02	3.84688E-01	9.16668E-02	-2.12951E+00	1.11059E+00
12	-5.59784E+00	2.30978E-01	-1.14241E-01	-2.79767E+00	-2.56919E+00	2.91441E+00	2.33216E-01	-5.60008E+00	2.91665E+00
13	4.47822E+00	-3.61731E-01	1.14241E-01	1.94400E+00	2.17248E+00	2.41997E+00	4.48091E+00	-3.64427E-01	2.42267E+00

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER						POINT					WRINKLING				
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
110	0	-60	-2.53E+0	-3.35E-1	-9.21E-1	115	-65	-2.54E+0	-5.52E-1	-1.18E+0	115	-65	-2.54E+0	-5.52E-1	-1.18E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPING			CONCENTRATION		
	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF	RF	RF	RF
(UP)	-567.6	533.4	585.20	-2310.8	1423.1	999.99	-160.7	482.08	156.61	145.13	185.23	
(LO)	-567.6	533.4	244.38	-2310.8	1423.1	690.89	-160.7	482.08				

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

POINT OF MAXIMUM Qx. COORDINATES x= 0.0, y= 184.0

INTERNAL LOADS : Nx = -1.0 Ny = 0.0 Nxy = 1.0 Mx = 0.0 My = 0.0 Mxy = 0.0 Qx = 2.35E-03 Qy = 0.00E+00

OUT-OF-PLANE DISPLACEMENT = 0.000

MIDDLE-PLANE DEFORMATIONS : (-1.695E-05 , 7.359E-06 , 2.934E-05)

LAYER	MICROSTRAINS IN FIB & MATRIX DIRECTIONS			MICROSTRAINS IN ELEMENT COORD SYSTEM		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY
1	9.964	-19.553	24.307	-16.948	7.359	29.517
2	-19.546	9.957	-24.307	-16.948	7.359	29.502
3	-16.948	7.359	29.487	-16.948	7.359	29.487
4	7.359	-16.948	-29.472	-16.948	7.359	29.472
5	-19.523	9.934	-24.307	-16.948	7.359	29.456
6	9.926	-19.515	24.307	-16.948	7.359	29.441
7	4.534	0.000	0.000	4.534	0.000	0.000
8	9.824	-19.413	24.307	-16.948	7.359	29.237
9	-19.406	9.817	-24.307	-16.948	7.359	29.222
10	7.359	-16.948	-29.207	-16.948	7.359	29.207
11	-16.948	7.359	29.192	-16.948	7.359	29.192
12	-19.383	9.794	-24.307	-16.948	7.359	29.176
13	9.786	-19.375	24.307	-16.948	7.359	29.161

LAYER	STRESSES IN FIBRE AND MATRIX DIRECTIONS			STRESSES IN ELEMENT COORDINATE SYSTEM			PRINCIPAL STRESSES		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY	MAJOR	MINOR	SHEAR
1	1.16557E+00	-1.66870E-01	1.14241E-01	3.85110E-01	6.13592E-01	6.66221E-01	1.17530E+00	-1.76594E-01	6.75945E-01
2	-2.37218E+00	4.12333E-02	-1.14241E-01	-1.27972E+00	-1.05123E+00	1.20671E+00	4.66291E-02	-2.37758E+00	1.21210E+00
3	-2.06075E+00	2.29141E-02	1.38588E-01	-2.06075E+00	2.29141E-02	1.38588E-01	3.20915E-02	-2.06993E+00	1.05101E+00
4	8.53229E-01	-1.48497E-01	-1.38516E-01	-1.48497E-01	8.53229E-01	1.38516E-01	8.72030E-01	-1.67298E-01	5.19664E-01
5	-2.36944E+00	4.10718E-02	-1.14241E-01	-1.27842E+00	-1.04994E+00	1.20525E+00	4.64739E-02	-2.37484E+00	1.21066E+00
6	1.16099E+00	-1.66601E-01	1.14241E-01	3.82956E-01	6.11438E-01	6.63797E-01	1.17075E+00	-1.76359E-01	6.73556E-01
7	2.18543E-04	0.00000E+00	0.00000E+00	2.18543E-04	0.00000E+00	0.00000E+00	2.18543E-04	0.00000E+00	0.00000E+00
8	1.14879E+00	-1.65883E-01	1.14241E-01	3.77213E-01	6.05695E-01	6.57337E-01	1.15864E+00	-1.75736E-01	6.67190E-01
9	-2.35540E+00	4.02462E-02	-1.14241E-01	-1.27182E+00	-1.04334E+00	1.19782E+00	4.56818E-02	-2.36084E+00	1.20326E+00
10	8.53229E-01	-1.48497E-01	-1.37272E-01	-1.48497E-01	8.53229E-01	1.37272E-01	8.71699E-01	-1.66967E-01	5.19333E-01
11	-2.06075E+00	2.29141E-02	1.37200E-01	-2.06075E+00	2.29141E-02	1.37200E-01	3.19093E-02	-2.06975E+00	1.05083E+00
12	-2.35265E+00	4.00846E-02	-1.14241E-01	-1.27053E+00	-1.04204E+00	1.19637E+00	4.55267E-02	-2.35810E+00	1.20181E+00
13	1.14421E+00	-1.65613E-01	1.14241E-01	3.75059E-01	6.03541E-01	6.54913E-01	1.15410E+00	-1.75503E-01	6.64803E-01

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER						POINT					WRINKLING				
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
50	-45	-70	-2.30E+0	-1.62E+0	1.93E+0	60	-76	-2.85E+0	-9.50E-1	1.65E+0	60	-76	-2.85E+0	-9.50E-1	1.65E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPING		CONCENTRATION		
	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF fiber	RF point	RF Wrin
(UP)	-567.6	533.4	506.76	-2310.8	1423.1	999.99	-160.7	482.08	134.48	124.93	152.90
(LO)	-567.6	533.4	509.54	-2310.8	1423.1	999.99	-160.7	482.08			

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

POINT OF MAXIMUM Qy. COORDINATES x= 376.5, y= 0.0

INTERNAL LOADS : Nx = -1.0 Ny = 0.0 Nxy = 1.0 Mx = 0.0 My = 0.0 Mxy = 0.0 Qx = 0.00E+00 Qy = 9.04E-02

OUT-OF-PLANE DISPLACEMENT = 0.000

MIDDLE-PLANE DEFORMATIONS : (-1.695E-05 , 7.359E-06 , 2.934E-05)

LAYER	MICROSTRAINS IN FIB & MATRIX DIRECTIONS			MICROSTRAINS IN ELEMENT COORD SYSTEM		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY
1	9.871	-19.459	24.307	-16.948	7.359	29.330
2	-19.460	9.871	-24.307	-16.948	7.359	29.331
3	-16.948	7.359	29.333	-16.948	7.359	29.333
4	7.359	-16.948	-29.334	-16.948	7.359	29.334
5	-19.462	9.873	-24.307	-16.948	7.359	29.336
6	9.874	-19.463	24.307	-16.948	7.359	29.337
7	0.000	348.968	0.000	0.000	348.968	0.000
8	9.876	-19.465	24.307	-16.948	7.359	29.341
9	-19.466	9.877	-24.307	-16.948	7.359	29.343
10	7.359	-16.948	-29.344	-16.948	7.359	29.344
11	-16.948	7.359	29.345	-16.948	7.359	29.345
12	-19.468	9.879	-24.307	-16.948	7.359	29.347
13	9.880	-19.469	24.307	-16.948	7.359	29.348

LAYER	STRESSES IN FIBRE AND MATRIX DIRECTIONS			STRESSES IN ELEMENT COORDINATE SYSTEM			PRINCIPAL STRESSES		
	NORMAL-1	NORMAL-2	SHEAR-12	NORMAL-X	NORMAL-Y	SHEAR-XY	MAJOR	MINOR	SHEAR
1	1.15434E+00	-1.66209E-01	1.14241E-01	3.79825E-01	6.08307E-01	6.60275E-01	1.16415E+00	-1.76019E-01	6.70085E-01
2	-2.36195E+00	4.06317E-02	-1.14241E-01	-1.27490E+00	-1.04642E+00	1.20129E+00	4.60517E-02	-2.36737E+00	1.20671E+00
3	-2.06075E+00	2.29141E-02	1.37865E-01	-2.06075E+00	2.29141E-02	1.37865E-01	3.19964E-02	-2.06984E+00	1.05092E+00
4	8.53229E-01	-1.48497E-01	-1.37872E-01	-1.48497E-01	8.53229E-01	1.37872E-01	8.71858E-01	-1.67126E-01	5.19492E-01
5	-2.36222E+00	4.06473E-02	-1.14241E-01	-1.27503E+00	-1.04654E+00	1.20143E+00	4.60666E-02	-2.36764E+00	1.20685E+00
6	1.15478E+00	-1.66235E-01	1.14241E-01	3.80033E-01	6.08515E-01	6.60509E-01	1.16459E+00	-1.76042E-01	6.70316E-01
7	0.00000E+00	8.41012E-03	0.00000E+00	0.00000E+00	8.41012E-03	0.00000E+00	0.00000E+00	8.41012E-03	0.00000E+00
8	1.15500E+00	-1.66248E-01	1.14241E-01	3.80136E-01	6.08618E-01	6.60625E-01	1.16481E+00	-1.76053E-01	6.70430E-01
9	-2.36262E+00	4.06707E-02	-1.14241E-01	-1.27521E+00	-1.04673E+00	1.20164E+00	4.60891E-02	-2.36804E+00	1.20706E+00
10	8.53229E-01	-1.48497E-01	-1.37917E-01	-1.48497E-01	8.53229E-01	1.37917E-01	8.71870E-01	-1.67138E-01	5.19504E-01
11	-2.06075E+00	2.29141E-02	1.37924E-01	-2.06075E+00	2.29141E-02	1.37924E-01	3.20041E-02	-2.06984E+00	1.05092E+00
12	-2.36288E+00	4.06863E-02	-1.14241E-01	-1.27534E+00	-1.04686E+00	1.20178E+00	4.61042E-02	-2.36830E+00	1.20720E+00
13	1.15544E+00	-1.66274E-01	1.14241E-01	3.80344E-01	6.08826E-01	6.60859E-01	1.16525E+00	-1.76076E-01	6.70661E-01

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER						POINT					WRINKLING				
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
50	-45	-70	-2.29E+0	-1.61E+0	1.92E+0	60	-75	-2.84E+0	-9.48E-1	1.64E+0	60	-75	-2.84E+0	-9.48E-1	1.64E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPING			CONCENTRATION		
	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF	RF	RF	RF
(UP)	-567.6	533.4	508.20	-2310.8	1423.1	999.99	-160.7	482.08	134.96	125.30	153.36	
(LO)	-567.6	533.4	508.09	-2310.8	1423.1	999.99	-160.7	482.08				

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

GENERAL BUCKLING ALLOWABLE LOADS

CASE	APPLIED FLOW			BUCKLING FLOW			NOX NOY		RF
	FX	FY	FX	FY	FX	FY			
1	-1.0	0.0	1.0	-207.2	0.0	207.2	2	2	207.15

MINIMUM GENERAL BUCKLING RF = 207.15 FOR LOAD CASE 1

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (ONLY WITH INPLANE LOADS)

CASE		FIBER			POINT			WRINKLING		
No	Ang Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
1	50 -45	-70	-2.29E+0	-1.61E+0	1.92E+0	60	-75	-2.84E+0	-9.48E-1	1.64E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (WHOLE PANEL ONLY WITH INPLANE LOADS)

CASE	GENERAL BUCKLING		WRINKLING			DIMPLING			SHEAR CRIMPLING		CONCENTRATION		
	No	RF	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF fiber	RF point	RF wrin
1	207.15	(UP) (LO)	-567.6 -567.6	533.4 533.4	508.15 508.15	-2310.8 -2310.8	1423.1 1423.1	999.99 999.99	-160.7 -160.7	482.08 482.08	134.94	125.29	153.34

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

SUMMARY OF RESERVE FACTORS FOR PANEL: panel_3

LOAD CASE No 1:

RF BUCKLING: 207.15

	INPLANE	
	FACE	RF
WRINKLING:	UP&LO	508.15
DIMPLING :	UP&LO	999.99
CRIMPLING:	UP&LO	482.08
CONC. Fib:	UPPER	134.94
CONC. Pnt:	UPPER	125.29
CONC. Wrk:	UPPER	153.34

	INPLANE+NORMAL LOADS			
	FACE	X	Y	RF
LOWER	376.50	184.00	228.97	
LOWER	0.00	0.00	690.89	
LOWER	376.50	184.00	94.64	
UPPER	376.50	184.00	92.20	
UPPER	376.50	184.00	90.35	
UPPER	0.00	184.00	152.90	

C.A.S.A. Criterium: YES

NO: LOWER	376.50	184.00
LOWER	376.50	184.00
BUCKL	0.00	0.00

ej_pansa_3.AUX

PANEL : panel_3 LOAD CASE : 1

Displacement and curvature on the panel centre and shear loads on the centre of the sides

ITER	NX	NY	Wc	Kmx	Kmy	Kmxy	Kx	Ky	Kxy	Qx	Qy
1	7	5	0.13	-3.70E-06	7.43E-06	4.87E-09	-1.14E-05	8.13E-06	2.90E-08	1.63E-01	1.03E-01
2	9	7	0.14	-2.06E-06	7.38E-06	5.93E-09	-3.62E-06	1.07E-05	3.88E-08	1.84E-02	8.95E-02
3	11	9	0.14	-5.60E-07	7.33E-06	3.81E-09	8.11E-06	1.93E-05	5.96E-09	1.44E-01	8.19E-02
4	13	11	0.14	-1.77E-06	7.41E-06	3.44E-09	-4.51E-06	1.24E-05	-2.16E-09	2.99E-01	9.10E-02
5	15	13	0.13	-2.84E-06	7.46E-06	6.01E-09	-2.06E-05	-9.79E-07	5.90E-08	1.61E-01	9.67E-02
6	16	14	0.14	-1.91E-06	7.39E-06	5.89E-09	-3.15E-06	1.07E-05	6.00E-08	2.35E-03	9.04E-02
7	16	14	0.14	-1.91E-06	7.39E-06	4.93E-09	-3.15E-06	1.07E-05	1.60E-08	2.35E-03	9.04E-02

Auxiliar information about the buckling calculation:

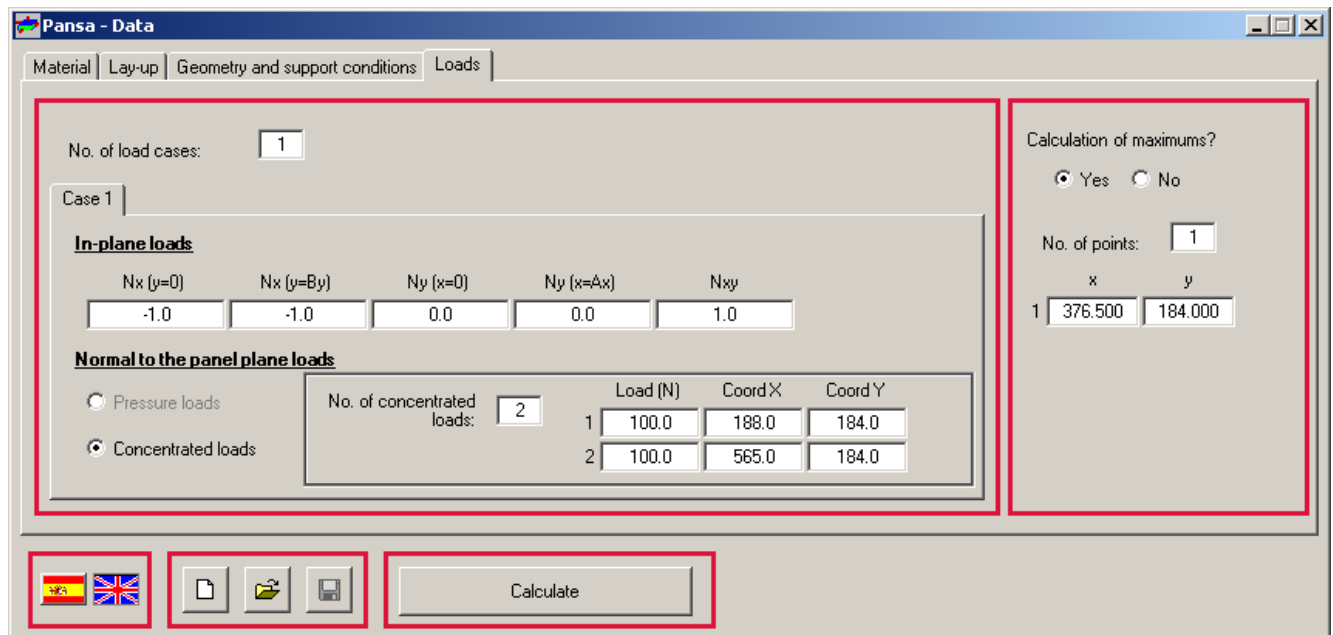
Number of iterations : 3

Relative difference : 0.21

ITER	NX	NY	FACTOR
1	4	3	210.048
2	6	5	207.596
3	8	7	207.154

BUCKLING MODE	RESERVE FACTOR	X DIRECTION SEMIWAVES	Y DIRECTION SEMIWAVES
1	207.154	2	2
2	208.786	3	1
3	228.910	4	4
4	235.505	4	2
5	252.933	6	2

If the PC version is used and the input file shown before is open, the program shows the following on the screen:



Pansa - Data

Material | Lay-up | Geometry and support conditions | **Loads**

No. of load cases:

Case 1

In-plane loads

Nx (y=0)	Nx (y=By)	Ny (x=0)	Ny (x=Ax)	Nxy
-1.0	-1.0	0.0	0.0	1.0

Normal to the panel plane loads

☐ Pressure loads

☒ Concentrated loads

No. of concentrated loads:

	Load (N)	Coord X	Coord Y
1	100.0	188.0	184.0
2	100.0	565.0	184.0

Calculation of maximums?
☒ Yes ☐ No

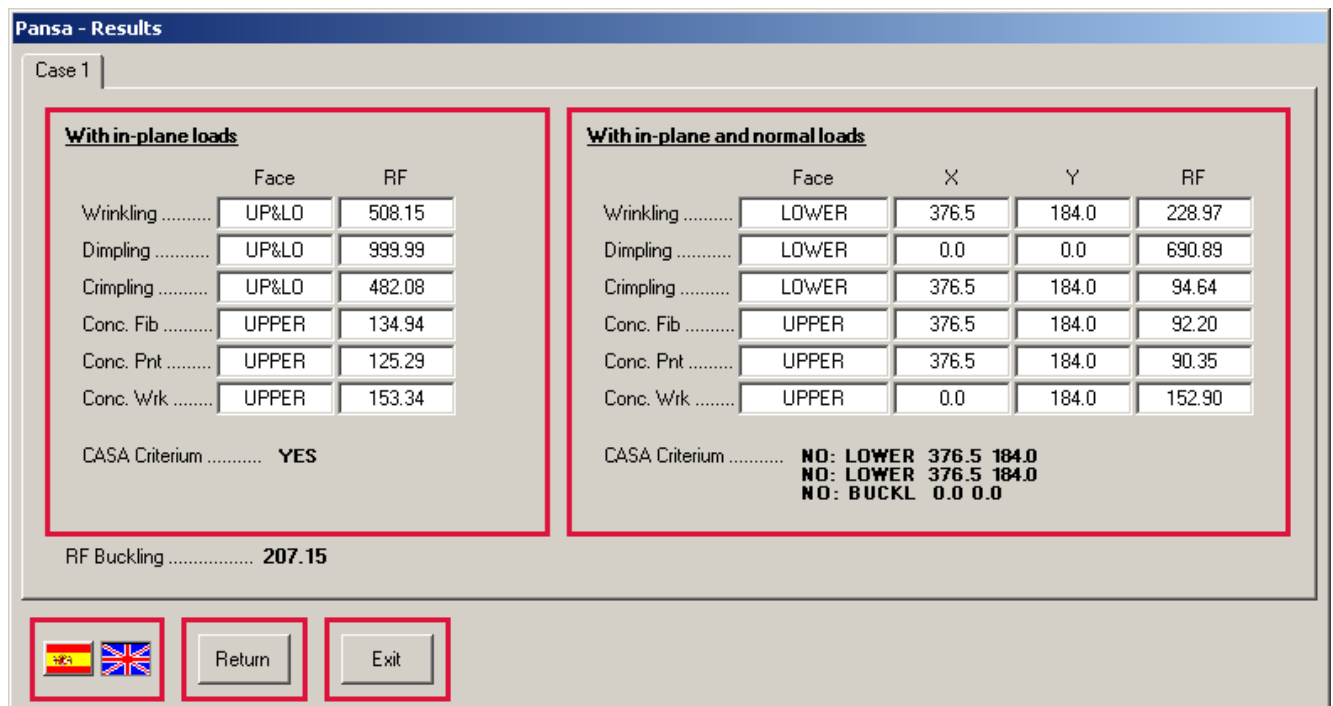
No. of points:

	x	y
1	376.500	184.000

Figure 8.5. Example 3 input data

If the 'Calculate' button is pressed the results are shown:



Pansa - Results

Case 1

With in-plane loads

	Face	RF
Wrinkling	UP&LO	508.15
Dimpling	UP&LO	999.99
Crimpling	UP&LO	482.08
Conc. Fib	UPPER	134.94
Conc. Pnt	UPPER	125.29
Conc. Wrk	UPPER	153.34

CASA Criterion YES

RF Buckling 207.15

With in-plane and normal loads

	Face	X	Y	RF
Wrinkling	LOWER	376.5	184.0	228.97
Dimpling	LOWER	0.0	0.0	690.89
Crimpling	LOWER	376.5	184.0	94.64
Conc. Fib	UPPER	376.5	184.0	92.20
Conc. Pnt	UPPER	376.5	184.0	90.35
Conc. Wrk	UPPER	0.0	184.0	152.90

CASA Criterion NO: LOWER 376.5 184.0
NO: LOWER 376.5 184.0
NO: BUCKL 0.0 0.0

Figure 8.6. Example 3 output results

8.4 Example 4

The fourth panel is rectangular, 753 mm length, 368 mm width and symmetrical. Faces thicknesses are 0.75 mm and the faces in-plane stiffness matrix is:

$$[A]_{mean\ values} = \begin{bmatrix} 36350.0 & 15783.8 & 0.0 \\ 15783.8 & 36350.0 & 0.0 \\ 0.0 & 0.0 & 17042.1 \end{bmatrix} \quad [A]_{allowable\ values} = \begin{bmatrix} 32650.9 & 13970.5 & 0.0 \\ 13970.5 & 32650.9 & 0.0 \\ 0.0 & 0.0 & 15305.4 \end{bmatrix}$$

And the allowable strains are:

$$\varepsilon_t = 12500E-06 \text{ mm/mm}$$

$$\varepsilon_c = 9500E-06 \text{ mm/mm}$$

The panel is simply supported and subjected to two load cases:

1. A compression flow on the x direction of 1 N/mm and a pressure z -negative direction with the value of 6881 N/m² on the points (0,0) and (753,368) and of 3440 N/m² on the points (753,0) and (0,368).
2. A positive shear flow of 1 N/mm and two normal loads of 100 N on points with coordinates $x = 188$ mm, $y = 184$ mm and $x = 565$ mm, $y = 184$ mm.

The properties of the core are:

$$E_c = 193 \text{ N/mm}^2 \quad (\text{Compression elastic modulus})$$

$$G_{xz} = 48.2 \text{ N/mm}^2 \quad (\text{Transverse shear modulus on the } xz \text{ plane})$$

$$G_{yz} = 24.1 \text{ N/mm}^2 \quad (\text{Transverse shear modulus on the } yz \text{ plane})$$

$$t = 10 \text{ mm.} \quad (\text{Core thickness})$$

$$s = 4.8 \text{ mm.} \quad (\text{Cell size})$$

Next, the data input file for this case is indicated.

ej_pansa_4.dat

```

! Materials
4
1,122000,10000,4700,.3,.125,0,12500E-6,9500E-6,109000,9000,4500
2,122000,10000,4700,.3,.125,45,12500E-6,9500E-6,109000,9000,4500
3,122000,10000,4700,.3,.125,-45,12500E-6,9500E-6,109000,9000,4500
4,122000,10000,4700,.3,.125,90,12500E-6,9500E-6,109000,9000,4500
! Laminate
1
.75
36350.8,15783.8,0.0,36350.8,0.0,17042.1
32650.9,13970.5,0.0,32650.9,0.0,15305.4,12500E-6,9500E-6
S
! Core properties
193.0,48.2,24.1,10.0,4.8
! Panel dimensions
753,368
! Support conditions
1,0.,0.
panel_4
! Load cases
2
-1.,-1.,0.,0.,0.
1,-6.881E-3,-3.440E-3,-3.440E-3,-6.881E-3,
0.,0.,0.,0.,1.
2,2,100,188,184,100,565,184
1,376.5,184
N

```

ej_pansa_4.RES

26-03-2008, Pansa 5.0

PANEL : panel_4

SANDWICH PANEL PROPERTIES

INPLANE STIFFNESS MATRIX (Mean Values)						INPLANE STIFFNESS MATRIX (B Values)					
UPPER FACE			LOWER FACE			UPPER FACE			LOWER FACE		
36350.8	15783.8	0.0	36350.8	15783.8	0.0	32650.9	13970.5	0.0	32650.9	13970.5	0.0
15783.8	36350.8	0.0	15783.8	36350.8	0.0	13970.5	32650.9	0.0	13970.5	32650.9	0.0
0.0	0.0	17042.1	0.0	0.0	17042.1	0.0	0.0	15305.4	0.0	0.0	15305.4
INPLANE ELASTIC MODULUS (Mean Values)						INPLANE ELASTIC MODULUS (B Values)					
Ex	Ey	Exy	Nux	Nuy		Ex	Ey	Exy			
UP&LOW FACES:	39329.80	39329.80	22722.80	0.43	0.43	35564.36	35564.36	20407.20			
						Ecc	Gxz	Gyz			
CORE PROPERTIES:						193.00	48.20	24.10			

SANDWICH STIFFNESS MATRIX (Mean Values)

INPLANE STIFFNESS MATRIX			COUPLING MATRIX			BENDING STIFFNESS MATRIX		
72701.6	31567.6	0.0	0.0	0.0	0.0	2100394.8	912007.7	0.0
31567.6	72701.6	0.0	0.0	0.0	0.0	912007.7	2100394.8	0.0
0.0	0.0	34084.2	0.0	0.0	0.0	0.0	0.0	984713.8
Ex	Ey	Exy	Nux	Nuy		Dx	Dy	Dxy
5130.0	5130.0	2963.8	0.43	0.43		1704393.9	1704393.9	984713.8

SANDWICH STIFFNESS MATRIX (B Values)

INPLANE STIFFNESS MATRIX			COUPLING MATRIX			BENDING STIFFNESS MATRIX		
65301.8	27941.0	0.0	0.0	0.0	0.0	1886609.9	807232.9	0.0
27941.0	65301.8	0.0	0.0	0.0	0.0	807232.9	1886609.9	0.0
0.0	0.0	30610.8	0.0	0.0	0.0	0.0	0.0	884365.2
Ex	Ey	Exy	Dx	Dy	Dxy			
4638.8	4638.8	2661.8	1541215.1	1541215.1	884365.2			

SANDWICH PANEL GEOMETRY

DIMENSION X	: 753.000
DIMENSION Y	: 368.000
UPPER FACE THICKNESS	: 0.750
CORE THICKNESS	: 10.000
LOWER FACE THICKNESS	: 0.750
TOTAL THICKNESS	: 11.500
CELL SIZE	: 4.800

BOUNDARY CONDITIONS : SIMPLY SUPPORTED

GENERALLY ORTOTROPIC PANEL

PANEL : panel_4

LOAD CASE : 1

INPLANE APPLIED FLOWS : FX = -1.0 FY = 0.0 FXY = 0.0

APPLIED PRESSURE LOAD : P (x= 0.0, y= 0.0) =-0.0069 P (x= 753.0, y= 0.0) =-0.0034
P (x= 753.0, y= 368.0) =-0.0034 P (x= 0.0, y= 368.0) =-0.0069

POINT : 1 WITH COORDINATES x= 376.5, y= 184.0

INTERNAL LOADS : Nx = -1.0 Ny = 0.0 Nxy = 0.0 Mx = -37.6 My = -64.4 Mxy = 0.0 Qx = 6.56E-02 Qy =-4.45E-08

OUT-OF-PLANE DISPLACEMENT = -0.673

MIDDLE-PLANE DEFORMATIONS : (-1.695E-05 , 7.360E-06 , 0.000E+00)

MEAN MICROSTRAINS				MEAN STRESSES		
FACE	NORMAL-X	NORMAL-Y	SHEAR-XY	NORMAL-X	NORMAL-Y	SHEAR-XY
UPPER	-47.386	-144.194	0.000	-5.33123E+00	-7.98597E+00	-3.63165E-08
CORE	126.564	0.000	0.000	6.10041E-03	-4.14409E-09	0.00000E+00
LOWER	13.484	158.914	0.000	3.99790E+00	7.98597E+00	3.63165E-08

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER						POINT					WRINKLING				
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
0	90	-411	0.00E+0	-1.62E+1	0.00E+0	0	-411	0.00E+0	-1.62E+1	0.00E+0	0	-411	0.00E+0	-1.62E+1	0.00E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPLING		CONCENTRATION		
	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF fiber	RF point	RF Wrin
(UP)	-416.4	535.6	52.14	-2366.6	1457.5	177.71	-160.7	20.12	23.11	23.11	28.09
(LO)	-454.1	535.6	999.99	-2366.6	1457.5	999.99	-160.7	999.99			

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

PANEL : panel_4

LOAD CASE : 2

INPLANE APPLIED FLOWS : FX = 0.0 FY = 0.0 FXY = 1.0

APPLIED CONCENTRATED LOAD : F (x= 188.0, y= 184.0) = 100.000
F (x= 565.0, y= 184.0) = 100.000

POINT : 1 WITH COORDINATES x= 376.5, y= 184.0

INTERNAL LOADS : Nx = 0.0 Ny = 0.0 Nxy = 1.0 Mx = 2.6 My = 13.7 Mxy = 0.0 Qx = 4.33E-08 Qy = 1.07E-08

OUT-OF-PLANE DISPLACEMENT = 0.137

MIDDLE-PLANE DEFORMATIONS : (0.000E+00 , 0.000E+00 , 2.934E-05)

MEAN MICROSTRAINS				MEAN STRESSES		
FACE	NORMAL-X	NORMAL-Y	SHEAR-XY	NORMAL-X	NORMAL-Y	SHEAR-XY
UPPER	-10.382	39.516	29.359	3.28418E-01	1.69676E+00	6.67120E-01
CORE	0.000	0.000	0.000	4.02881E-09	9.92624E-10	0.00000E+00
LOWER	10.382	-39.516	29.319	-3.28418E-01	-1.69676E+00	6.66213E-01

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER						POINT					WRINKLING				
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
165	90	122	3.56E-1	4.96E+0	1.33E+0	160	126	6.61E-1	4.99E+0	1.82E+0	70	-39	-1.56E+0	-2.07E-1	5.68E-1

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPLING		CONCENTRATION		
	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF fiber	RF point	RF Wrin
(UP)	-454.1	535.6	802.78	-2366.6	1457.5	999.99	-160.7	999.99	102.22	98.57	323.01
(LO)	-453.0	535.6	242.67	-2366.6	1457.5	948.79	-160.7	94.69			

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

GENERAL BUCKLING ALLOWABLE LOADS

CASE	APPLIED FLOW			BUCKLING FLOW			NOX NOY		RF
	FX	FY	FX	FY	FX	FY			
1	-1.0	0.0	0.0	-335.9	0.0	0.0	3	1	335.85
2	0.0	0.0	1.0	0.0	0.0	324.7	2	2	324.72

MINIMUM GENERAL BUCKLING RF = 324.72 FOR LOAD CASE 2

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (ONLY WITH INPLANE LOADS)

CASE			FIBER				POINT				WRINKLING					
No	Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
1	90	0	-45	-1.79E+0	0.00E+0	0.00E+0	90	-45	-1.79E+0	0.00E+0	0.00E+0	90	-45	-1.79E+0	0.00E+0	0.00E+0
2	45	-45	-52	-1.45E+0	-1.45E+0	1.45E+0	45	-52	-1.45E+0	-1.45E+0	1.45E+0	45	-52	-1.45E+0	-1.45E+0	1.45E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (WHOLE PANEL ONLY WITH INPLANE LOADS)

CASE	GENERAL BUCKLING		WRINKLING			DIMPLING			SHEAR CRIMPING		CONCENTRATION		
	No	RF	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF fiber	RF point	RF wrin
1	335.85	(UP)	-572.2	535.6	858.27	-2366.6	1457.5	999.99	-321.3	482.00	208.20	208.20	318.84
		(LO)	-572.2	535.6	858.27	-2366.6	1457.5	999.99	-321.3	482.00			
2	324.72	(UP)	-454.1	535.6	803.33	-2366.6	1457.5	999.99	-160.7	999.99	179.45	179.45	210.30
		(LO)	-454.1	535.6	803.33	-2366.6	1457.5	999.99	-160.7	999.99			

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

SUMMARY OF RESERVE FACTORS FOR PANEL: panel_4

LOAD CASE No 1:

RF BUCKLING: 335.85

	INPLANE		INPLANE+NORMAL LOADS			
	FACE	RF	FACE	X	Y	RF
WRINKLING:	UP&LO	858.27	UPPER	376.50	184.00	52.14
DIMPLING :	UP&LO	999.99	UPPER	376.50	184.00	177.71
CRIMPLING:	UP&LO	482.00	UPPER	376.50	184.00	20.12
CONC. Fib:	UPPER	208.20	UPPER	376.50	184.00	23.11
CONC. Pnt:	UPPER	208.20	UPPER	376.50	184.00	23.11
CONC. Wrk:	UPPER	318.84	UPPER	376.50	184.00	28.09

C.A.S.A. Criterium: YES

NO: UPPER 376.50 184.00

LOAD CASE No 2:

RF BUCKLING: 324.72

	INPLANE		INPLANE+NORMAL LOADS			
	FACE	RF	FACE	X	Y	RF
WRINKLING:	UP&LO	803.33	LOWER	376.50	184.00	242.67
DIMPLING :	UP&LO	999.99	LOWER	376.50	184.00	948.79
CRIMPLING:	UP&LO	999.99	LOWER	376.50	184.00	94.69
CONC. Fib:	UPPER	179.45	UPPER	376.50	184.00	102.22
CONC. Pnt:	UPPER	179.45	UPPER	376.50	184.00	98.57
CONC. Wrk:	UPPER	210.30	UPPER	376.50	184.00	323.01

C.A.S.A. Criterium: YES

NO: LOWER 376.50 184.00

ej_pansa_4.AUX

PANEL : panel_4 LOAD CASE : 1

Displacement and curvature on the panel centre and shear loads on the centre of the sides

ITER	NX	NY	Wc	Kmx	Kmy	Kmxy	Kx	Ky	Kxy	Qx	Qy
1	7	5	-0.67	-6.18E-06	-2.78E-05	-2.99E-13	-8.60E-06	-4.04E-05	-5.19E-13	-7.48E-01	-7.64E-01
2	7	5	-0.67	-5.66E-06	-2.82E-05	-2.97E-13	-6.44E-06	-4.55E-05	-4.81E-13	-8.18E-01	-7.81E-01

Auxiliar information about the buckling calculation:

Number of iterations : 2

Relative difference : 0.00

ITER	NX	NY	FACTOR
1	4	3	335.852
2	6	5	335.852

BUCKLING MODE	RESERVE FACTOR	X DIRECTION SEMIWAVES	Y DIRECTION SEMIWAVES
1	335.852	3	1
2	352.761	2	1
3	358.310	4	1
4	388.552	5	1
5	417.077	6	1

PANEL : panel_4 LOAD CASE : 2

Displacement and curvature on the panel centre and shear loads on the centre of the sides

ITER	NX	NY	Wc	Kmx	Kmy	Kmxy	Kx	Ky	Kxy	Qx	Qy
1	7	5	0.13	-3.74E-06	7.39E-06	3.71E-09	-1.14E-05	8.09E-06	2.79E-08	1.62E-01	1.03E-01
2	9	7	0.14	-2.09E-06	7.34E-06	4.82E-09	-3.64E-06	1.06E-05	3.78E-08	1.76E-02	8.94E-02
3	11	9	0.14	-5.72E-07	7.28E-06	3.09E-09	8.09E-06	1.92E-05	5.21E-09	1.43E-01	8.18E-02
4	13	11	0.14	-1.79E-06	7.37E-06	2.80E-09	-4.52E-06	1.24E-05	-2.61E-09	2.98E-01	9.09E-02
5	15	13	0.13	-2.87E-06	7.42E-06	4.42E-09	-2.06E-05	-1.01E-06	5.78E-08	1.60E-01	9.66E-02
6	16	14	0.14	-1.93E-06	7.35E-06	4.41E-09	-3.16E-06	1.07E-05	5.89E-08	1.18E-03	9.03E-02
7	16	14	0.14	-1.93E-06	7.35E-06	3.71E-09	-3.16E-06	1.07E-05	1.51E-08	1.17E-03	9.03E-02

Auxiliar information about the buckling calculation:

Number of iterations : 3

Relative difference : 0.48

ITER	NX	NY	FACTOR
1	4	3	340.969
2	6	5	326.284
3	8	7	324.720

BUCKLING MODE	RESERVE FACTOR	X DIRECTION SEMIWAVES	Y DIRECTION SEMIWAVES
1	324.720	2	2
2	327.593	3	3
3	359.252	4	2
4	360.864	4	4
5	387.289	5	3

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

Pansa - Data

Material | Lay-up | Geometry and support conditions | Loads

To define the laminate

☐ using the stacking sequence ☒ using stiffness matrix ☐ using elastic moduli

☒ Symmetrical

Upper face

Thickness: 0.750 $\mu_e t$ 12500 $\mu_e c$ 9500

Matrix A (mean values)

36350.8	15783.8	0.0
15783.8	36350.8	0.0
0.0	0.0	17042.1

Matrix A (allowables)

32650.9	13970.5	0.0
13970.5	32650.9	0.0
0.0	0.0	15305.4

Lower face

Thickness: 0.750 $\mu_e t$ 12500 $\mu_e c$ 9500

Matrix A (mean values)

36350.8	15783.8	0.0
15783.8	36350.8	0.0
0.0	0.0	17042.1

Matrix A (allowables)

32650.9	13970.5	0.0
13970.5	32650.9	0.0
0.0	0.0	15305.4

Calculate

Figure 8.7. Example 4 input data

If the 'Calculate' button is pressed the results are shown:

Pansa - Results

Case 1 | Case 2

With in-plane loads

	Face	RF
Wrinkling	UP&LO	858.27
Dimpling	UP&LO	999.99
Crimpling	UP&LO	482.00
Conc. Fib	UPPER	208.20
Conc. Pnt	UPPER	208.20
Conc. Wrk	UPPER	318.84

CASA Criterium **YES**

RF Buckling **335.85**

With in-plane and normal loads

	Face	X	Y	RF
Wrinkling	UPPER	376.5	184.0	52.14
Dimpling	UPPER	376.5	184.0	177.71
Crimpling	UPPER	376.5	184.0	20.12
Conc. Fib	UPPER	376.5	184.0	23.11
Conc. Pnt	UPPER	376.5	184.0	23.11
Conc. Wrk	UPPER	376.5	184.0	28.09

CASA Criterium **NO: UPPER 376.5 184.0**

Return Exit

Figure 8.8. Example 4 output results

8.5 Example 5

The fifth panel has the characteristic data entry of a metallic faces sandwich. It has the same dimensions as the previous panel and the same core. Faces are equal and orthotropic, their thickness is 0.75 mm and their properties are:

Average properties:

$$\begin{aligned} E_x &= 39329.9 \text{ N/mm}^2 & E_y &= 39329.9 \text{ N/mm}^2 \\ G &= 22722.7 \text{ N/mm}^2 & \mu &= 0.43 \end{aligned}$$

Allowable properties, (B values):

$$\begin{aligned} E_x &= 35564.3 \text{ N/mm}^2 & E_y &= 35564.3 \text{ N/mm}^2 & G &= 20407.2 \text{ N/mm}^2 \\ \epsilon_t &= 12500\text{E-}06 \text{ mm/mm} & \epsilon_c &= 9500\text{E-}06 \text{ mm/mm} \end{aligned}$$

It is simply supported and subject to the same load cases as the previous panel.

Next, the data input file for this case is indicated.

ej_pansa_5.dat

```
! Materials
4
1,122000,10000,4700,.3,.125,0,12500E-6,9500E-6,109000,9000,4500
2,122000,10000,4700,.3,.125,45,12500E-6,9500E-6,109000,9000,4500
3,122000,10000,4700,.3,.125,-45,12500E-6,9500E-6,109000,9000,4500
4,122000,10000,4700,.3,.125,90,12500E-6,9500E-6,109000,9000,4500
! Laminate
2
.75
39329.9,39329.9,22722.7,0.43
35564.3,35564.3,20407.2,12500E-6,9500E-6
S
! Core properties
193.0,48.2,24.1,10.0,4.8
! Panel dimensions
753,368
! Support conditions
1,0.,0.
panel_5
! Load cases
2
-1.,-1.,0.,0.,0.
1,-6.881E-3,-3.440E-3,-3.440E-3,-6.881E-3,
0.,0.,0.,0.,1.
2,2,100,188,184,100,565,184
1,376.5,184
N
```

ej_pansa_5.RES

26-03-2008, Pansa 5.0

PANEL : panel_5

SANDWICH PANEL PROPERTIES

INPLANE STIFFNESS MATRIX (Mean Values)						INPLANE STIFFNESS MATRIX (B Values)					
UPPER FACE			LOWER FACE			UPPER FACE			LOWER FACE		
36188.7	15561.1	0.0	36188.7	15561.1	0.0	32723.9	14071.3	0.0	32723.9	14071.3	0.0
15561.1	36188.7	0.0	15561.1	36188.7	0.0	14071.3	32723.9	0.0	14071.3	32723.9	0.0
0.0	0.0	17042.0	0.0	0.0	17042.0	0.0	0.0	15305.4	0.0	0.0	15305.4
INPLANE ELASTIC MODULUS (Mean Values)						INPLANE ELASTIC MODULUS (B Values)					
		Ex	Ey	Exy	Nux	Nuy			Ex	Ey	Exy
UP&LOW FACES:		39329.90	39329.90	22722.70	0.43	0.43			35564.30	35564.30	20407.20
						Ecc	Gxz	Gyz			
CORE PROPERTIES:						193.00	48.20	24.10			

SANDWICH STIFFNESS MATRIX (Mean Values)

INPLANE STIFFNESS MATRIX			COUPLING MATRIX			BENDING STIFFNESS MATRIX		
72377.4	31122.3	0.0	0.0	0.0	0.0	2094422.0	900601.6	0.0
31122.3	72377.4	0.0	0.0	0.0	0.0	900601.6	2094422.0	0.0
0.0	0.0	34084.0	0.0	0.0	0.0	0.0	0.0	986307.1
	Ex	Ey	Exy	Nux	Nuy	Dx	Dy	Dxy
	5130.0	5130.0	2963.8	0.43	0.43	1707163.2	1707163.2	986307.1

SANDWICH STIFFNESS MATRIX (B Values)

INPLANE STIFFNESS MATRIX			COUPLING MATRIX			BENDING STIFFNESS MATRIX		
65447.7	28142.5	0.0	0.0	0.0	0.0	1893894.0	814374.5	0.0
28142.5	65447.7	0.0	0.0	0.0	0.0	814374.5	1893894.0	0.0
0.0	0.0	30610.8	0.0	0.0	0.0	0.0	0.0	885800.0
		Ex	Ey	Exy		Dx	Dy	Dxy
		4638.8	4638.8	2661.8		1543712.9	1543712.9	885800.0

SANDWICH PANEL GEOMETRY

DIMENSION X	:	753.000
DIMENSION Y	:	368.000
UPPER FACE THICKNESS	:	0.750
CORE THICKNESS	:	10.000
LOWER FACE THICKNESS	:	0.750
TOTAL THICKNESS	:	11.500
CELL SIZE	:	4.800

BOUNDARY CONDITIONS : SIMPLY SUPPORTED

GENERALLY ORTOTROPIC PANEL

PANEL : panel_5 LOAD CASE : 1

INPLANE APPLIED FLOWS : FX = -1.0 FY = 0.0 FXY = 0.0

APPLIED PRESSURE LOAD : P (x= 0.0, y= 0.0) =-0.0069 P (x= 753.0, y= 0.0) =-0.0034
P (x= 753.0, y= 368.0) =-0.0034 P (x= 0.0, y= 368.0) =-0.0069

POINT : 1 WITH COORDINATES x= 376.5, y= 184.0

INTERNAL LOADS : Nx = -1.0 Ny = 0.0 Nxy = 0.0 Mx = -37.4 My = -64.3 Mxy = 0.0 Qx = 6.56E-02 Qy =-4.45E-08

OUT-OF-PLANE DISPLACEMENT = -0.674

MIDDLE-PLANE DEFORMATIONS : (-1.695E-05 , 7.289E-06 , 0.000E+00)

MEAN MICROSTRAINS				MEAN STRESSES		
FACE	NORMAL-X	NORMAL-Y	SHEAR-XY	NORMAL-X	NORMAL-Y	SHEAR-XY
UPPER	-47.460	-144.700	0.000	-5.29228E+00	-7.96671E+00	-3.64054E-08
CORE	126.557	0.000	0.000	6.10003E-03	-4.14374E-09	0.00000E+00
LOWER	13.559	159.277	0.000	3.95895E+00	7.96671E+00	3.64054E-08

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER						POINT					WRINKLING				
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
0	90	-411	0.00E+0	-1.62E+1	0.00E+0	0	-411	0.00E+0	-1.62E+1	0.00E+0	0	-411	0.00E+0	-1.62E+1	0.00E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPLING		CONCENTRATION		
	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF fiber	RF point	RF Wrin
(UP)	-416.8	535.7	52.32	-2356.0	1457.5	177.69	-160.7	20.17	23.10	23.10	28.08
(LO)	-454.1	535.7	999.99	-2356.0	1457.5	999.99	-160.7	999.99			

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

PANEL : panel_5

LOAD CASE : 2

INPLANE APPLIED FLOWS : FX = 0.0 FY = 0.0 FXY = 1.0

APPLIED CONCENTRATED LOAD : F (x= 188.0, y= 184.0) = 100.000
F (x= 565.0, y= 184.0) = 100.000

POINT : 1 WITH COORDINATES x= 376.5, y= 184.0

INTERNAL LOADS : Nx = 0.0 Ny = 0.0 Nxy = 1.0 Mx = 2.6 My = 13.7 Mxy = 0.0 Qx = 4.34E-08 Qy = 1.07E-08

OUT-OF-PLANE DISPLACEMENT = 0.137

MIDDLE-PLANE DEFORMATIONS : (0.000E+00 , 0.000E+00 , 2.934E-05)

MEAN MICROSTRAINS				MEAN STRESSES		
FACE	NORMAL-X	NORMAL-Y	SHEAR-XY	NORMAL-X	NORMAL-Y	SHEAR-XY
UPPER	-10.396	39.615	29.359	3.20315E-01	1.69579E+00	6.67121E-01
CORE	0.000	0.000	0.000	4.04079E-09	9.92991E-10	0.00000E+00
LOWER	10.396	-39.615	29.319	-3.20315E-01	-1.69579E+00	6.66213E-01

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (NORMAL AND INPLANE LOADS)

FIBER						POINT					WRINKLING				
Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
165	90	122	3.57E-1	4.97E+0	1.33E+0	160	127	6.62E-1	4.99E+0	1.82E+0	70	-40	-1.58E+0	-2.09E-1	5.75E-1

RESERVE FACTORS AND ALLOWABLES SUMMARY (LOCAL RESULTS WITH NORMAL AND INPLANE LOADS)

	WRINKLING			DIMPLING			SHEAR CRIMPLING		CONCENTRATION		
	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF fiber	RF point	RF Wrin
(UP)	-454.1	535.7	802.99	-2356.0	1457.5	999.99	-160.7	999.99	102.04	98.39	319.37
(LO)	-453.1	535.7	242.83	-2356.0	1457.5	948.80	-160.7	94.74			

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

GENERAL BUCKLING ALLOWABLE LOADS

CASE	APPLIED FLOW			BUCKLING FLOW			NOX NOY		RF
	FX	FY	FX	FY	FX	FY			
1	-1.0	0.0	0.0	-336.4	0.0	0.0	3	1	336.45
2	0.0	0.0	1.0	0.0	0.0	325.1	2	2	325.07

MINIMUM GENERAL BUCKLING RF = 325.07 FOR LOAD CASE 2

MAXIMUM STRESSES AND STRAINS DUE TO CONCENTRATION (ONLY WITH INPLANE LOADS)

CASE			FIBER				POINT				WRINKLING					
No	Ang	Ori	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY	Ang	Strain	SigX	SigY	SigXY
1	90	0	-45	-1.80E+0	0.00E+0	0.00E+0	90	-45	-1.80E+0	0.00E+0	0.00E+0	90	-45	-1.80E+0	0.00E+0	0.00E+0
2	45	-45	-53	-1.45E+0	-1.45E+0	1.45E+0	45	-53	-1.45E+0	-1.45E+0	1.45E+0	45	-53	-1.45E+0	-1.45E+0	1.45E+0

RESERVE FACTORS AND ALLOWABLES SUMMARY (WHOLE PANEL ONLY WITH INPLANE LOADS)

CASE	GENERAL BUCKLING		WRINKLING			DIMPLING			SHEAR CRIMPING		CONCENTRATION		
	No	RF	Fcw	Fsw	RF	Fcd	Fsd	RF	Fsc	RF	RF fiber	RF point	RF wrin
1	336.45	(UP) (LO)	-572.2 -572.2	535.7 535.7	858.27 858.27	-2356.0 -2356.0	1457.5 1457.5	999.99 999.99	-321.3 -321.3	482.00 482.00	208.01 208.01	208.01 208.01	318.54 318.54
2	325.07	(UP) (LO)	-454.1 -454.1	535.7 535.7	803.54 803.54	-2356.0 -2356.0	1457.5 1457.5	999.99 999.99	-160.7 -160.7	999.99 999.99	179.07 179.07	179.07 179.07	210.50 210.50

NOTE: 999.99 VALUE MEANS THAT VARIABLE HAS A VALUE HIGHER THAN THE VALUE SHOWN

SUMMARY OF RESERVE FACTORS FOR PANEL: panel_5

LOAD CASE No 1:

RF BUCKLING: 336.45

	INPLANE		INPLANE+NORMAL LOADS			
	FACE	RF	FACE	X	Y	RF
WRINKLING:	UP&LO	858.27	UPPER	376.50	184.00	52.32
DIMPLING :	UP&LO	999.99	UPPER	376.50	184.00	177.69
CRIMPLING:	UP&LO	482.00	UPPER	376.50	184.00	20.17
CONC. Fib:	UPPER	208.01	UPPER	376.50	184.00	23.10
CONC. Pnt:	UPPER	208.01	UPPER	376.50	184.00	23.10
CONC. Wrk:	UPPER	318.54	UPPER	376.50	184.00	28.08

C.A.S.A. Criterium: YES

NO: UPPER 376.50 184.00

LOAD CASE No 2:

RF BUCKLING: 325.07

	INPLANE		INPLANE+NORMAL LOADS			
	FACE	RF	FACE	X	Y	RF
WRINKLING:	UP&LO	803.54	LOWER	376.50	184.00	242.83
DIMPLING :	UP&LO	999.99	LOWER	376.50	184.00	948.80
CRIMPLING:	UP&LO	999.99	LOWER	376.50	184.00	94.74
CONC. Fib:	UPPER	179.07	UPPER	376.50	184.00	102.04
CONC. Pnt:	UPPER	179.07	UPPER	376.50	184.00	98.39
CONC. Wrk:	UPPER	210.50	UPPER	376.50	184.00	319.37

C.A.S.A. Criterium: YES

NO: LOWER 376.50 184.00

ej_pansa_5.AUX

PANEL : panel_5 LOAD CASE : 1

Displacement and curvature on the panel centre and shear loads on the centre of the sides

ITER	NX	NY	Wc	Kmx	Kmy	Kmxy	Kx	Ky	Kxy	Qx	Qy
1	7	5	-0.67	-6.20E-06	-2.79E-05	-3.00E-13	-8.61E-06	-4.05E-05	-5.20E-13	-7.48E-01	-7.64E-01
2	7	5	-0.67	-5.68E-06	-2.83E-05	-2.98E-13	-6.45E-06	-4.56E-05	-4.81E-13	-8.18E-01	-7.81E-01

Auxiliar information about the buckling calculation:

Number of iterations : 2

Relative difference : 0.00

ITER	NX	NY	FACTOR
1	4	3	336.447
2	6	5	336.447

BUCKLING MODE	RESERVE FACTOR	X DIRECTION SEMI WAVES	Y DIRECTION SEMI WAVES
1	336.447	3	1
2	353.501	2	1
3	358.844	4	1
4	389.037	5	1
5	417.511	6	1

PANEL : panel_5 LOAD CASE : 2

Displacement and curvature on the panel centre and shear loads on the centre of the sides

ITER	NX	NY	Wc	Kmx	Kmy	Kmxy	Kx	Ky	Kxy	Qx	Qy
1	7	5	0.13	-3.75E-06	7.41E-06	3.72E-09	-1.14E-05	8.11E-06	2.79E-08	1.62E-01	1.03E-01
2	9	7	0.14	-2.09E-06	7.35E-06	4.83E-09	-3.64E-06	1.06E-05	3.78E-08	1.76E-02	8.94E-02
3	11	9	0.14	-5.73E-07	7.30E-06	3.09E-09	8.09E-06	1.93E-05	5.22E-09	1.43E-01	8.18E-02
4	13	11	0.14	-1.79E-06	7.38E-06	2.80E-09	-4.52E-06	1.24E-05	-2.60E-09	2.98E-01	9.09E-02
5	15	13	0.13	-2.87E-06	7.44E-06	4.42E-09	-2.06E-05	-9.95E-07	5.78E-08	1.60E-01	9.66E-02
6	16	14	0.14	-1.93E-06	7.37E-06	4.41E-09	-3.16E-06	1.07E-05	5.89E-08	1.20E-03	9.04E-02
7	16	14	0.14	-1.93E-06	7.37E-06	3.72E-09	-3.16E-06	1.07E-05	1.52E-08	1.20E-03	9.04E-02

Auxiliar information about the buckling calculation:

Number of iterations : 3

Relative difference : 0.48

ITER	NX	NY	FACTOR
1	4	3	341.405
2	6	5	326.637
3	8	7	325.072

BUCKLING MODE	RESERVE FACTOR	X DIRECTION SEMI WAVES	Y DIRECTION SEMI WAVES
1	325.072	2	2
2	327.943	3	3
3	359.478	4	2
4	361.066	4	4
5	387.485	5	3

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

Figure 8.9. Example 5 input data

If the 'Calculate' button is pressed the results are shown:

	Face	RF
Wrinkling	UP&LO	803.54
Dimpling	UP&LO	999.99
Crimpling	UP&LO	999.99
Conc. Fib	UPPER	179.07
Conc. Pnt	UPPER	179.07
Conc. Wrk	UPPER	210.50

CASA Criterion YES

RF Buckling 325.07

	Face	X	Y	RF
Wrinkling	LOWER	376.5	184.0	242.83
Dimpling	LOWER	376.5	184.0	948.80
Crimpling	LOWER	376.5	184.0	94.74
Conc. Fib	UPPER	376.5	184.0	102.04
Conc. Pnt	UPPER	376.5	184.0	98.39
Conc. Wrk	UPPER	376.5	184.0	319.37

CASA Criterion NO: LOWER 376.5 184.0

Figure 8.10. Example 5 output results

8.6 Example 6

The sixth panel is like the first and it is subjected to the same load cases. The difference is that in that case no points of calculations are given, so only the buckling analysis is carried out.

The data input file is the following:

ej_pansa_6.dat

```
! Materials
4
1,122000,10000,4700,.3,.125,0,12500E-6,9500E-6,109000,9000,4500
2,122000,10000,4700,.3,.125,45,12500E-6,9500E-6,109000,9000,4500
3,122000,10000,4700,.3,.125,-45,12500E-6,9500E-6,109000,9000,4500
4,122000,10000,4700,.3,.125,90,12500E-6,9500E-6,109000,9000,4500
! Laminate
0
6
2,3,1,4,3,2
S
! Core properties
193.0,48.2,24.1,10.0,4.8
! Panel dimensions
753,368
! Support conditions
1,0.,0.
panel_1
! Load cases
2
-1.,-1.,0.,0.,0.
1,-6.881E-3,-6.881E-3,-6.881E-3,-6.881E-3
0.,0.,0.,0.,1.
0
i Points of analysis
0
N
```

ej_pansa_6.RES

26-03-2008, Pansa 5.0

PANEL : panel_6

SANDWICH PANEL PROPERTIES

LAYUP

NUM LAY	LAY THICKN.	M. PLANE DISTAN.	(Mean Values)			POISSON RATIO	LAY ORIE.	ALLOW. TENSION	MICROSTRAINS COMPRESSION	(B Values)		
			E1 (Gxz)	E2 (Gyz)	G (Ecc)					ALLOWABLE	MODULUS	
										E1	E2	G
1	0.125	5.688	122000.	10000.	4700.	0.30	45	12500.	9500.	109000.	9000.	4500.
2	0.125	5.562	122000.	10000.	4700.	0.30	-45	12500.	9500.	109000.	9000.	4500.
3	0.125	5.438	122000.	10000.	4700.	0.30	0	12500.	9500.	109000.	9000.	4500.
4	0.125	5.312	122000.	10000.	4700.	0.30	90	12500.	9500.	109000.	9000.	4500.
5	0.125	5.188	122000.	10000.	4700.	0.30	-45	12500.	9500.	109000.	9000.	4500.
6	0.125	5.062	122000.	10000.	4700.	0.30	45	12500.	9500.	109000.	9000.	4500.
7	10.0000	0.0000	0.4820E+02	0.2410E+02	0.1930E+03							
8	0.125	-5.062	122000.	10000.	4700.	0.30	45	12500.	9500.	109000.	9000.	4500.
9	0.125	-5.188	122000.	10000.	4700.	0.30	-45	12500.	9500.	109000.	9000.	4500.
10	0.125	-5.312	122000.	10000.	4700.	0.30	90	12500.	9500.	109000.	9000.	4500.
11	0.125	-5.438	122000.	10000.	4700.	0.30	0	12500.	9500.	109000.	9000.	4500.
12	0.125	-5.562	122000.	10000.	4700.	0.30	-45	12500.	9500.	109000.	9000.	4500.
13	0.125	-5.688	122000.	10000.	4700.	0.30	45	12500.	9500.	109000.	9000.	4500.

INPLANE STIFFNESS MATRIX (Mean Values)

UPPER FACE

36350.8	15783.8	0.0	36350.8	15783.8	0.0
15783.8	36350.8	0.0	15783.8	36350.8	0.0
0.0	0.0	17042.1	0.0	0.0	17042.1

LOWER FACE

INPLANE ELASTIC MODULUS (Mean Values)

	Ex	Ey	Exy	Nux	Nuy
UP&LOW FACES:	38401.92	38401.97	22722.74	0.43	0.43

SANDWICH STIFFNESS MATRIX (Mean Values)

INPLANE STIFFNESS MATRIX

72711.6	31567.5	0.0
31567.5	72701.7	0.0
0.0	0.0	34084.1

COUPLING MATRIX

0.0	0.0	0.0
0.0	0.0	0.0
0.0	0.0	0.0

BENDING STIFFNESS MATRIX

2122275.5	914049.0	881.5
914049.0	2084289.5	881.0
881.5	881.0	986871.9

Ex	Ey	Exy	Nux	Nuy	Dx	Dy	Dxy
5130.9	5130.2	2963.8	0.43	0.43	1721426.4	1690615.0	986871.9

SANDWICH STIFFNESS MATRIX (B Values)

INPLANE STIFFNESS MATRIX

65311.8	27941.1	0.0
27941.1	65301.9	0.0
0.0	0.0	30610.8

COUPLING MATRIX

0.0	0.0	0.0
0.0	0.0	0.0
0.0	0.0	0.0

BENDING STIFFNESS MATRIX

1906179.5	809041.9	787.1
809041.9	1872253.0	786.6
787.1	786.6	886295.8

Ex	Ey	Exy	Dx	Dy	Dxy
4639.7	4639.0	2661.8	1556574.6	1528870.5	886295.8

SANDWICH PANEL GEOMETRY

DIMENSION X	: 753.000
DIMENSION Y	: 368.000
UPPER FACE THICKNESS	: 0.750
CORE THICKNESS	: 10.000
LOWER FACE THICKNESS	: 0.750
TOTAL THICKNESS	: 11.500
CELL SIZE	: 4.800

BOUNDARY CONDITIONS : SIMPLY SUPPORTED

NO GENERALLY ORTOTROPIC PANEL

GENERAL BUCKLING ALLOWABLE LOADS

CASE	APPLIED FLOW			BUCKLING FLOW			NOX	NOY	RF
	FX	FY	FXY	FX	FY	FXY			
1	-1.0	0.0	0.0	-337.0	0.0	0.0	3	1	336.97
2	0.0	0.0	1.0	0.0	0.0	324.9	2	2	324.95
MINIMUM GENERAL BUCKLING RF = 324.95 FOR LOAD CASE							2		

ej_pansa_6.AUX

PANEL : panel_6 LOAD CASE : 1

Auxiliar information about the buckling calculation:

Number of iterations : 2

Relative difference : 0.00

ITER	NX	NY	FACTOR
1	4	3	336.968
2	6	5	336.969

BUCKLING MODE	RESERVE FACTOR	X DIRECTION SEMIWAVES	Y DIRECTION SEMIWAVES
1	336.969	3	1
2	353.423	2	1
3	359.596	4	1
4	389.838	5	1
5	418.270	6	1

PANEL : panel_6 LOAD CASE : 2

Auxiliar information about the buckling calculation:

Number of iterations : 3

Relative difference : 0.48

ITER	NX	NY	FACTOR
1	4	3	341.199
2	6	5	326.511
3	8	7	324.945

BUCKLING MODE	RESERVE FACTOR	X DIRECTION SEMIWAVES	Y DIRECTION SEMIWAVES
1	324.945	2	2
2	327.784	3	3
3	359.451	4	2
4	361.116	4	4
5	387.418	5	3

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

Pansa - Data

Material | Lay-up | Geometry and support conditions | **Loads**

No. of load cases:

Case 1 | Case 2

In-plane loads

Nx (y=0)	Nx (y=By)	Ny (x=0)	Ny (x=Ax)	Nxy
-1.0	-1.0	0.0	0.0	0.0

Normal to the panel plane loads

☒ Pressure loads

Pressure at x=0 y=0	-0.00688	Pressure at x=Ax y=By ...	-0.00688
Pressure at x=Ax y=0	-0.00688	Pressure at x=0 y=By	-0.00688

☐ Concentrated loads

Calculation of maximums?
☐ Yes ☒ No

No. of points:

Calculate

Figure 8.11. Example 6 input data

If the 'Calculate' button is pressed the results are shown:

Pansa - Results

Case 1 | Case 2

BUCKLING ANALYSIS

	Fx (y=0)	Fx (y=By)	Fy (x=0)	Fy (x=Ax)	Fxy
Applied Loads	-1.0	-1.0	0.0	0.0	0.0
Buckling Loads	-337.0	-337.0	0.0	0.0	0.0

RF buckling 336.97

Return **Exit**

Figure 8.12. Example 6 output results